Testing What Is Usually Assumed: Proportional Hazards and Linearity in a Counting-Process Model of Customer Churn

Evidence from 6,000 Users of a Vehicle Diagnostics App

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Master's Degree Project in

Data Science, Statistics, and Decision Analysis

Autumn term 2026

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Abstract

Introduction Churn studies usually fit the Cox model without testing its two assumptions: proportional hazards and linear covariate effects. This thesis tests both on app activity logs from OBDeleven, a vehicle diagnostics company, where churn is defined as 160 days without activity.

Research questions Do the data reject proportional hazards or linearity, and how large are any departures? Does relaxing linearity, or both assumptions together, improve discrimination and calibration for customers not used in fitting?

Method A retrospective analysis of the app's activity logs covers 6,000 private customers, followed in 14-day intervals with covariates updated at every interval. Six models are compared, from a linear Cox model through spline, grouped complementary log-log and time-varying Cox models to gradient-boosted trees, using Schoenfeld residual tests, likelihood ratio tests and five-fold cross-validation grouped by customer.

Results The global test does not reject proportional hazards, but the effect of the one-click app proportion covariate may reverse over follow-up. Linearity is rejected under both the Cox and the complementary log-log link ($p < 10^{-8}$), and tied event times do not explain it. Out of fold, every cross-validated model reaches an area under the ROC curve of about 0.68. Splines raise it by only 0.002, while gradient-boosted trees raise it by 0.005 and have the calibration slope closest to 1.

Discussion The assumptions can fail with little effect on prediction but still change how the hazard ratios read, so they should be tested and the size of any departure reported rather than assumed. The coefficients are associations, and they apply only to customers who stayed at least 56 days.

Acknowledgements

Antonis Matakos read draft after draft of this thesis and kept pushing me to cut it down and say things plainly. It is a much better thesis because of that.

OBDeleven gave me the data and, early on, helped me work out which question was worth answering for them.

Abbreviations

Abbreviation	Description
AUC	Area under the ROC curve
CI	Confidence interval
clog-log	Complementary log-log
CV	Coefficient of variation
EPV	Events per variable
HHI	Herfindahl–Hirschman index
HR	Hazard ratio
KM	Kaplan–Meier
PH	Proportional hazards
ROC	Receiver operating characteristic
RQ	Research question
SD	Standard deviation
SHAP	Shapley additive explanations

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1

Introduction

Customer churn is the loss of customers who stop using a product or service. Predicting which customers are likely to leave, and when, lets a company target retention before the revenue is lost. Churn prediction is often framed as binary classification, asking whether a customer churns within a fixed horizon such as 90 days. This framing struggles with customers who are still active when data collection ends. Their churn time is not observed, only that it lies beyond the end of the data, and such observations are called censored. Whether a censored customer churns by the horizon is unknown, so a classifier must drop them or assign a guessed label (Section 3.1).

Survival analysis models how long a customer stays before churning, instead of whether they churn by a fixed date. It estimates the churn rate among customers still active at each point in time, so a customer counts for every day they were observed. A customer observed for 60 days cannot be labelled for a 90-day horizon, but still contributes 60 days without churn to a survival model.

A survival model works with the hazard, the rate at which customers who are still active churn at a given time. Covariates are the customer characteristics used to explain this rate, such as the number of sessions in the last 28 days. The Cox model, the most common model in the churn studies reviewed in Section 2.5, links them to the hazard as

$$\lambda(t \mid x) = \lambda_0(t) \exp(\beta_1 x_1 + \cdots + \beta_p x_p),$$

where $\lambda_0(t)$ is a baseline hazard shared by all customers and left free to change over time. For two customers who differ by one unit in x_k and are equal otherwise, $\lambda_0(t)$ cancels, and the ratio of their hazards is $\exp(\beta_k)$, the hazard ratio. It is read much as an odds ratio is in logistic regression. A hazard ratio of 0.8 for sessions means that each extra session lowers the hazard by 20%. It does not mean that the probability of churning falls by 20%.

The Cox model assumes proportional hazards and linearity. Proportional hazards means the hazard ratio is the same at every point of follow-up, so an extra session has the same effect in a customer's second month as in the second year. Linearity means the log of the hazard changes in a straight line with each covariate, so going from 1 to 2 sessions has the same effect as going from 20 to 21. Each covariate can be summarised by one hazard ratio only when both hold.

When an effect changes over follow-up, the fitted hazard ratio is a weighted average of the effects at different times and describes no single period [38]. When an effect is not linear, one ratio per unit overstates the effect over part of the covariate's range and understates it over the rest. Tests of these assumptions often have low power [5]. A test that does not reject therefore does not show that an assumption holds, so the size of any departure should be reported as well.

In the Cox churn studies reviewed in Section 2.5, these assumptions are usually taken for granted. Only one of them tests proportional hazards formally [24], none tests linearity, and none checks whether a violation affects prediction. A review of Cox studies in other fields finds the same gap for proportional hazards [22].

This thesis tests both assumptions in a churn model whose covariates change over time, and measures whether relaxing them improves prediction. The work falls within applied data science, in survival modelling of customer churn.

RQ1. Do the data reject proportional hazards or linearity in the Cox model, and how large are any departures?

RQ2. Does relaxing linearity, or both assumptions together, improve discrimination and calibration for customers not used in fitting?

Discrimination measures whether customers who churn receive higher predicted risks than those who stay. Calibration measures whether the predicted risks match the churn rates actually observed.

The data are app activity logs from 6,000 customers. The logs contain no cancellation events, because a customer who leaves simply stops appearing in them. A customer is therefore treated as churned after 160 days without activity, and the churn is dated at their last active day (Section 3.3). Each customer's history is cut into 14-day intervals, and each interval becomes one row of data. A row records the customer's activity up to the end of its interval, such as the number of sessions in the last 28 days. Its outcome is whether the customer churns in the following 14 days.

The logs come from OBDeleven, a vehicle diagnostics company based in Kaunas, Lithuania. Its mobile app connects to a car's control units through a Bluetooth device in the OBD2 port, and lets users read and clear fault codes, view live data and change vehicle settings, either by coding them manually or with preset one-click apps [27]. Basic diagnostics are free, while coding and one-click apps are sold as yearly subscriptions or as credits, so a customer who stops using the app stops generating revenue. The app serves both private owners and workshops. This thesis studies only private owners, called the personal segment below (Section 3.2).

The thesis baseline is a Cox model in which every covariate enters as a straight line. Scaled Schoenfeld residuals test proportional hazards on this model, since they show whether a covariate's effect drifts over follow-up. A time-varying version then measures how large any drift is. Spline versions let each covariate's effect bend, and likelihood ratio tests compare them with the straight-line fits to test linearity. A discrete-time version of the Cox model repeats the linearity test, so that churn times shared by many customers can be ruled out as the cause. Gradient-boosted trees drop both assumptions and serve as a benchmark. Together these make six models, each differing from another in one respect (Section 3.7.1).

Five-fold cross-validation compares the models on prediction, except the time-varying version, which only measures the size of the drift. Each customer is kept in a single fold, so no model is scored on customers it was fitted on (Section 3.7.2).

This thesis contributes to the churn literature in three respects. First, it tests proportional hazards and linearity separately, in a churn model whose covariates change over time. Second, it measures whether relaxing linearity, alone or together with proportional hazards, improves predictions for customers not used in fitting. Third, it builds the churn event and the customer segment directly from raw, unlabelled activity logs and states each step.

Only each customer's first churn is modelled. All results apply to customers who stayed for at least the first 56 days. The coefficients show associations with churn, not causes.

The remaining chapters are organised as follows.

Chapter	Content
2	Survival analysis background and churn literature
3	Data pipeline and models
4	Assumption tests and out-of-fold comparison
5	Findings and limitations

2

Extended Background

Throughout this chapter, $i = 1, \ldots, n$ indexes subjects and $j = 1, \ldots, J$ indexes intervals. Table A.1 summarises the notation, and Appendix A provides the formulae and derivations behind the results below. The chapter moves from the basic quantities and censoring to the models used, the design issues specific to churn data, the two assumptions tested, and the churn studies that leave the gap this thesis addresses.

2.1 Foundations of survival analysis

2.1.1 Basic quantities

Survival analysis models the time until an event. The density $f(t)$, the distribution function $F(t)$, the survival function $S(t)$ and the continuous hazard $\lambda(t)$ (Appendix A.2) follow Kalbfleisch and Prentice [18, ch. 1]. Each answers a different question about the same event time: $F(t)$ and $S(t)$ how many have churned or remain by t, $f(t)$ when churn tends to happen, and $\lambda(t)$ the risk of churning now for a customer still active, which is the quantity hazard regression models estimate. The continuous hazard is a *rate*, bounded below by zero and unbounded above, and can be read as a probability only approximately, and only over a short interval. The discrete hazard $h(t)$ is a *probability*, confined to $[0, 1]$, and the separate symbol keeps the two quantities apart.

2.1.2 Censoring and the likelihood

Survival analysis exists as a separate field because of censoring. A customer still active at the cutoff is right-censored, since only a lower bound on the churn time is known [33]. Survival methods use this partial information instead of dropping the customer.

An often-overlooked assumption is that censoring is uninformative, meaning that the reasons why a subject stops being observed must not influence their

churn risk [36, ch. 9]. This does not require random censoring. Administrative censoring at a fixed cutoff is deterministic, yet uninformative, because the calendar date is unrelated to churn risk. If censoring is related to churn risk, the hazard estimate is biased (Eq. A.11). Kalbfleisch and Prentice [18, ch. 6] give the formal conditions under which censoring can be left out of the likelihood without losing information.

Every model below is estimated by maximising a likelihood, and censored subjects enter the estimation instead of being discarded. The likelihood combines both the density of failing and the probability of no event at time t, and can be written in terms of the hazard function alone (Appendix A.3). That hazard form is the starting point for both the Cox partial likelihood (Section 2.2.1) and the person-period likelihood (Section 2.2.2).

2.2 Hazard regression models

2.2.1 The Cox model

The Cox model factors the hazard into a baseline shared by all subjects and a subject-specific multiplier [6]. Assume for now one row per subject, with a covariate vector x_i fixed at the time origin:

$$\lambda(t \mid x_i) = \lambda_0(t) \exp(x_i' \beta), \tag{2.1}$$

where $\lambda_0(t)$ is the baseline hazard, the hazard of a subject with $x_i = 0$, and β is a vector of regression coefficients (non-time-varying for the time being).

The model is multiplicative on the hazard and therefore additive on the log hazard, so a unit increase in a covariate scales the hazard by e^{β} at every t. The hazard ratio between any two subjects does not depend on t (Appendix A.4). This is the proportional hazards assumption. Equivalently, β does not vary with t; a coefficient written $\beta(t)$ would make the ratio time-dependent.

The baseline $\lambda_0(t)$ is left unspecified, which makes the model semiparametric. As a result, the Cox model estimates covariate effects without assuming how churn risk changes over time. Because $\lambda_0(t)$ cancels in the ratio of two subjects' hazards, $\exp(\beta_k)$ gives the hazard ratio for a one-unit increase in covariate k, holding the other covariates fixed. These properties explain why it is the standard choice in survival analysis. Appendix A.8 sets out other relative risk functions whose hazard ratios depend on covariate values, and the additive model of Aalen [1].

Because $\lambda_0(t)$ is left unspecified, the coefficients cannot be estimated from the full likelihood. Cox [6] instead proposed a partial likelihood that uses only the order in which events occur. At each event time t_i, the contribution is the probability that the event happened to subject i rather than to another subject in the risk set $R(t_i)$,

$$L_i(\beta) = \frac{\lambda_0(t_i)\exp(\beta' x_i)}{\sum_{j \in R(t_i)} \lambda_0(t_i)\exp(\beta' x_j)} = \frac{\exp(\beta' x_i)}{\sum_{j \in R(t_i)} \exp(\beta' x_j)}. \tag{2.2}$$

The baseline cancels, so β can be estimated without specifying $\lambda_0(t)$. Each contribution is a probability, although it is built from a ratio of hazards, which are rates (Appendix A.4).

The same construction extends to covariates that change over time. The partial likelihood compares subjects only at each event time, so it suffices to use each subject's covariate values at that moment. The subject who had the event is compared with the current values of everyone else still at risk [43].

The partial likelihood treats time as continuous, so it assumes that no two subjects have the same event time [24]. The exact method, the Breslow approximation and the Efron approximation handle ties, and they agree when ties are few. As the share of ties grows, Breslow's approximation becomes less accurate than Efron's, which is therefore recommended when the exact method is too expensive [24].

2.2.2 Grouped and discrete time

This section gives the discrete-time counterpart of the Cox model.

The discrete hazard h_j is the conditional probability of the event occurring within $(t_{j-1}, t_j]$ given survival to its start [45, §3.3.2]. With the covariate vector held constant within each interval [43], the proportional hazards form (2.1) gives (Appendix A.5.2):

$$\log\big(-\log(1-h_j)\big) = \gamma_j + x'\beta, \tag{2.3}$$

where γ_j is the log of the baseline cumulative hazard over interval j. This is the grouped-data version of the proportional hazards model [32, 45]. Here h_j is a probability rather than a rate, bounded within $[0, 1]$, and (2.3) is precisely a generalised linear model with a complementary log-log link. It can therefore be fitted with standard binary regression software, with γ_j entering

as an interval-specific intercept [36, 40]. Importantly, β is the same β as in the continuous-time model (2.1).

A unit increase in the kth covariate multiplies the accumulated hazard by e^{β_k} (Appendix A.5.2), holding the others fixed, the same interpretation β carries in (2.1).

The probability of surviving to the end of an interval is the product of the conditional survival probabilities $1 - h_j$ of that interval and all earlier ones (Appendix A.5.3). In the person-period format, each subject contributes one row for every interval in which they are at risk, and d_{ij} indicates whether the event occurred in that interval. The resulting likelihood has the same form as that of a binomial model in which the d_{ij} are independent observations with probabilities h_{ij} that depend on the covariates x_{ij} [40]. Appendix A.5.4 gives the derivation.

The Kaplan–Meier estimator [19] estimates $S(t)$ without any assumption about the shape of the hazard, but it allows no covariates (Appendix A.6). The Cox model and the grouped model keep this freedom for the baseline and add the covariates.

The same person-period data can also be fitted with a logistic link (Eq. A.28). The clog-log form follows from grouping a continuous process, while the logistic form treats time as genuinely discrete. The two give similar estimates when intervals are short [45, §3.2].

2.2.3 Counting process and time-varying effects

The counting process formulation is what allows the Cox model to be fitted on interval rows with covariates that change over time [4]. It replaces each subject's single survival time with two functions of t. Let $N_i(t)$ count the events observed for subject i over $[0, t]$, and let $Y_i(t)$ equal one when subject i is under observation and event-free just before t, and zero otherwise. These two functions are exactly what the interval rows hold. Subject i contributes one row per interval, and the boundaries $(t_{j-1}, t_j]$ give $Y_i(t)$ while the event indicator gives $N_i(t)$.

The model specifies the counting-process intensity, which is the baseline rate scaled by the subject's covariates as they stand at t, and zero when the subject is not at risk (Appendix A.9). With covariates fixed, $x_i(t) = x_i$, this is the Cox model (2.1). The difference is that $x_i(t)$ may change while the subject is under observation. A customer should appear in the risk set only for periods in

which a churn would have shown up in the data. Before entry or after censoring, no churn could be recorded, so no row exists for those periods [43, §4.1].

The difference between a subject's observed and expected number of events is a martingale, and its central limit theorem gives the large-sample distribution of the Cox score and of the Schoenfeld residuals behind the test in Section 2.4 [3]. Delayed entry is handled by setting $Y_i(t)$ to zero until the subject joins, and censoring does not disturb the estimation as long as it is uninformative (Section 2.1.2) [3].

A time-varying covariate means $x_i(t)$ changes over time while its effect β stays the same, and interval rows allow this. A time-varying coefficient means the effect $\beta(t)$ itself changes over time. That breaks proportional hazards and has nothing to do with how the data are built [49]. The two are separate things, and mixing them up is a common error discussed in Section 2.5.

Time-varying covariates are either external or internal [18, §6.3]. An external covariate changes regardless of whether the subject survives, such as age or calendar time. An internal covariate is generated by the subject, so it can only be measured while they are still there.

Both types enter the model in the same way [18, §6.3]. An internal covariate, however, cannot be projected forward, because a customer's future activity exists only if they stay. A forecast over a longer horizon therefore needs a landmark, a time point at which the covariates are frozen [16].

2.3 Design choices

2.3.1 Continuous or discrete time

Researchers often choose between discrete and continuous time without justification [20].

Tutz and Schmid [45, Fig. 3.6] compare three models on the same simulated data with five covariates. These are the continuous Cox model with exact partial likelihood, the discrete logit model (which the authors call continuation-ratio), and the grouped proportional hazards model. The interval length is varied from 1 to 360 units. With fine intervals, the Cox estimates vary the least, but the grouped model has the smallest root mean squared error at every interval length. This is because the grouped model is correctly specified however wide the intervals are, while the Cox model has to approximate the ties.

As intervals widen, the Cox and logit models lose accuracy, and the grouped model is barely affected.

Kim [20] finds that logit and clog-log models diverge only when hazards are high, and that on empirical data logit and Cox models differ in the significance of only one predictor.

2.3.2 Survival biases

Immortal-time bias is often overlooked [39]. It happens when a subject is counted as exposed over a period in which they could not have churned, simply because they had to survive that period to become exposed. It can enter in two ways. The first is a covariate summarised over the whole follow-up, such as a maximum, mean, last value or count. The second is assigning subjects to groups based on something that can happen only to those who stay long enough for it to occur. Appendix A.10 gives a constructed example with ten subjects.

End-point bias, as described by Meža, Strle, and Meža [26], artificially inflates the survival function. A subject who joins ten days before the study ends, when churn requires forty days of inactivity, cannot churn in the time left. The subject is labelled a survivor by construction, not by behaviour, and as more such late joiners accumulate, the estimated survival is driven towards 1 (Appendix B.3).

2.4 Proportional hazards and linearity

The Cox model makes two separate claims, and they are usually treated as one.

Proportional hazards means β does not change over time, so the hazard ratio between two covariate values is the same early and late in follow-up (Section 2.2.1). *Linearity* means the log hazard is $x'\beta$, so a one-unit change in a covariate has the same effect regardless of its current value. The two can fail separately. A model can have proportional hazards and the wrong shape, or the right shape and effects that change over time. A model that does not assume proportional hazards is also not automatically a model of time-varying effects. Table 2.1 shows what each model assumes is constant over time. Only the Cox model with $\beta(t)$ and the Aalen additive model are built to estimate how effects change over time.

Specification	Held constant across time	Requires PH
Cox PH [6]	hazard ratio	Yes
Grouped PH, clog-log [32]	interval cumulative hazard ratio	Yes
Discrete logistic [45, §3.2]	odds ratio	No
Discrete probit [45, §3.2]	nothing	No
Cox with $\beta(t)$ [49]	nothing	No
Aalen additive [1]	nothing	No
Accelerated failure time [18, ch. 2]	time scaling	No
Kaplan–Meier [19]	(non-parametric)	No

Table 2.1: Whether common specifications rely on the proportional hazards assumption, and what each holds constant over time.

Both assumptions are testable. Proportional hazards is tested with scaled Schoenfeld residuals [11], which are a function of time under the alternative and flat under the null. Linearity can be assessed with martingale residuals [44] or by fitting the covariate as a smooth term and comparing linear and spline models [13].

A restricted cubic spline lets a covariate's effect curve by joining cubic pieces at a set of knots, while keeping the curve straight beyond the outer knots, where data are sparse [9]. With knots $t_1 < \cdots < t_K$, a covariate x enters the log hazard as

$$f(x) = \beta_1 x + \sum_{j=1}^{K-2} \beta_{j+1}\, s_j(x), \tag{2.4}$$

$$s_j(x) = (x - t_j)_+^3 - \frac{t_K - t_j}{t_K - t_{K-1}}\,(x - t_{K-1})_+^3 + \frac{t_{K-1} - t_j}{t_K - t_{K-1}}\,(x - t_K)_+^3,$$

where $(u)_+ = \max(u, 0)$ [15, §2.4.5]. If $\beta_2 = \cdots = \beta_{K-1} = 0$, the effect is linear, so a likelihood ratio test of those coefficients tests linearity.

The Schoenfeld test compares residuals against a transformed time scale, and the researcher chooses the transform, usually identity, log, Kaplan–Meier or rank. Park and Hendry [28] show that researchers often choose this transform without justification, and it can change the test's conclusion. They recommend choosing it based on the censoring level and whether outlying survival times exist, with the rank transform performing best as censoring increases.

The test's power is also low, even with many events [5]. With time-varying covariates, power depends on how the assumption is violated, which is only known in simulation, so high power cannot be expected in practice [12]. A non-significant result can therefore mean either that proportional hazards holds or that the test could not detect the violation.

The Schoenfeld test has two implementations. The traditional version uses a single covariance matrix of the covariates across all event times, while the newer version recalculates it at each event time. Metzger [25] finds that the newer implementation rejects proportional hazards far more often than it should when covariates are correlated. The implementation used should therefore be reported with the result.

Some authors question whether the test is worth running at all. Stensrud and Hernán [38] argue that hazards are rarely proportional in practice, so testing is unnecessary and the fitted hazard ratio can be read as a weighted average of the hazard ratios at each time. Sjölander and Dickman [37] add that no assumption holds exactly, so with a large enough sample every test rejects, and with a small one none do. The useful question is whether the departure is large enough to matter, which a p-value does not show. Its size can be read from how much the hazard ratio changes between early and late follow-up once the effect is allowed to vary.

2.5 Survival analysis in the churn literature

The churn studies below mostly use Cox models but rarely check their assumptions.

Two studies already use covariates that change over follow-up. Van den Poel and Larivière [46] fit a Cox model with time-varying covariates to customers of a financial-services company and say that plots of scaled Schoenfeld residuals support proportional hazards, but do not show them. Gür Ali and Arıtürk [14] stack each customer's observations from several periods, score them on the following five months, and use Cox regression only as a benchmark.

Kimitei et al. [21] note that proportional hazards is often assumed without testing it and say they run PH diagnostics, but report no result. They call average utility usage a time-varying factor the Cox model cannot analyse and switch to an additive model. It is unclear whether they mean a time-varying covariate, which the Cox model handles, or a time-varying coefficient, which violates proportional hazards and which they never test.

Meža, Strle, and Meža [26] state that both assumptions fail but report no test of either. They describe their features as non-linear, heavy-tailed and full of interactions, yet fit no spline and run no proportional hazards test. Their partial dependence and ICE curves come from the tree models, so they show what those models learned, not whether a linear Cox term would have been enough.

Their study covers only 1,074 users. Its covariates are computed once, from sessions before a single cutoff, so change over time enters only as fixed trend and volatility summaries. Its top feature accumulates over weekly bins and so grows with observation time, the cumulative-covariate route to immortal-time bias (Section 2.3.2). When they residualise that feature on observation time, the Cox C-index drops from 0.830 to 0.718.

Wong [48] enters the number of handset changes as a linear term. The 39.9 per cent lower churn reported for two changes therefore equals the hazard ratio for one change, 0.775, squared. It follows from the linear term rather than being a separate estimate. Both this and the rate-plan effect are also described as a lower chance of churning, which reads a hazard ratio as a probability (Section 2.1.1).

In Periáñez et al. [31], one covariate, lifetime, is the number of days from registration to churn, and two others are calculated from it. This introduces direct outcome leakage, so the reported performance does not represent prospective prediction. The model also cannot be applied to players who have not yet churned.

Only Masarifoglu and Buyuklu [24] test proportional hazards formally, none tests linearity, and none combines covariates updated over follow-up with both tests and scoring on held-out customers (Table 2.2).

Study	Covariates	PH test	Linearity test	Scored on
Van den Poel and Larivière [46]	time-varying	residual plots, not shown	no	none reported
Gür Ali and Arıtürk [14]	stacked periods	no	no	later-period holdout
Wong [48]	per-customer totals	no	no	none reported
Masarifoglu and Buyuklu [24]	fixed	yes	no	none reported
Periáñez et al. [31]	fixed per player	no	n/a	bootstrap, leaked outcome
Rahman et al. [34]	fixed, categorical	not reported	no	fitting data
Kimitei et al. [21]	monthly, binned	claimed, not reported	no (binned)	80/20 split
Meža, Strle, and Meža [26]	fixed at a cutoff	no	no	5-fold cross-validation

Table 2.2: Covariates, assumption tests and scoring in the reviewed churn studies.

3

Method

This chapter describes how the raw activity logs become the modelling data, in the order the steps run, and then the models fitted to them. The order matters, because every step that removes customers changes the population the estimates describe. Appendix B holds the derivations.

3.1 Data and design

This thesis is empirical research, since it describes, explains and predicts churn rather than designing an artefact [17, p. 1]. It is a retrospective, observational and longitudinal analysis of the behavioural logs of a vehicle diagnostics app. An experiment could show what causes churn, but customers cannot be randomly assigned to behave one way or another [8, pp. 66, 75–77]. The coefficients are therefore associations with churn, not causal effects. A questionnaire would record what customers say rather than what they do, and would miss those who had already left [8, pp. 17, 170–171].

The unit of analysis is the customer. Each customer is followed from the first recorded activity, and the follow-up is cut into 14-day intervals (Section 3.4). The logs record no cancellation, so churn is inferred from inactivity (Section 3.3).

The covariates describe the customer's own activity, such as sessions and scans. Such covariates are called internal, because they exist only while the customer is still there [18, §6.3]. A forecast for the months ahead would need the customer's future activity, which is unknown and itself depends on whether they churn. The model therefore predicts only the risk of churning in the next interval, from the activity so far.

Many customers are still active when the data end, so their churn time is censored. A classifier at a fixed horizon τ cannot label a customer censored before τ [35], and dropping them over-represents the customers who churn [30] (Appendix A.7). Its coefficients also hold only for that τ. A survival model keeps

censored customers up to their last known status, and its coefficients act on the hazard, so they do not depend on a horizon.

The data come from three tables: activity, vehicle and user (Figure 3.1). Vehicles are right-joined to activity on both `user_id` and `vehicle_id`, because the same `vehicle_id` can appear under several users. The right join keeps every activity row, including those with no matching vehicle. Workshops scan many vehicles, so more of their activity has no match. An inner join would drop that activity and make workshops look like private owners under the rule in Section 3.2. The only rows lost are vehicles with no activity, which are assumed to be unused. Customers missing from the vehicle table are removed, because no vehicle covariates can be built for them. The user table only adds the registration date.

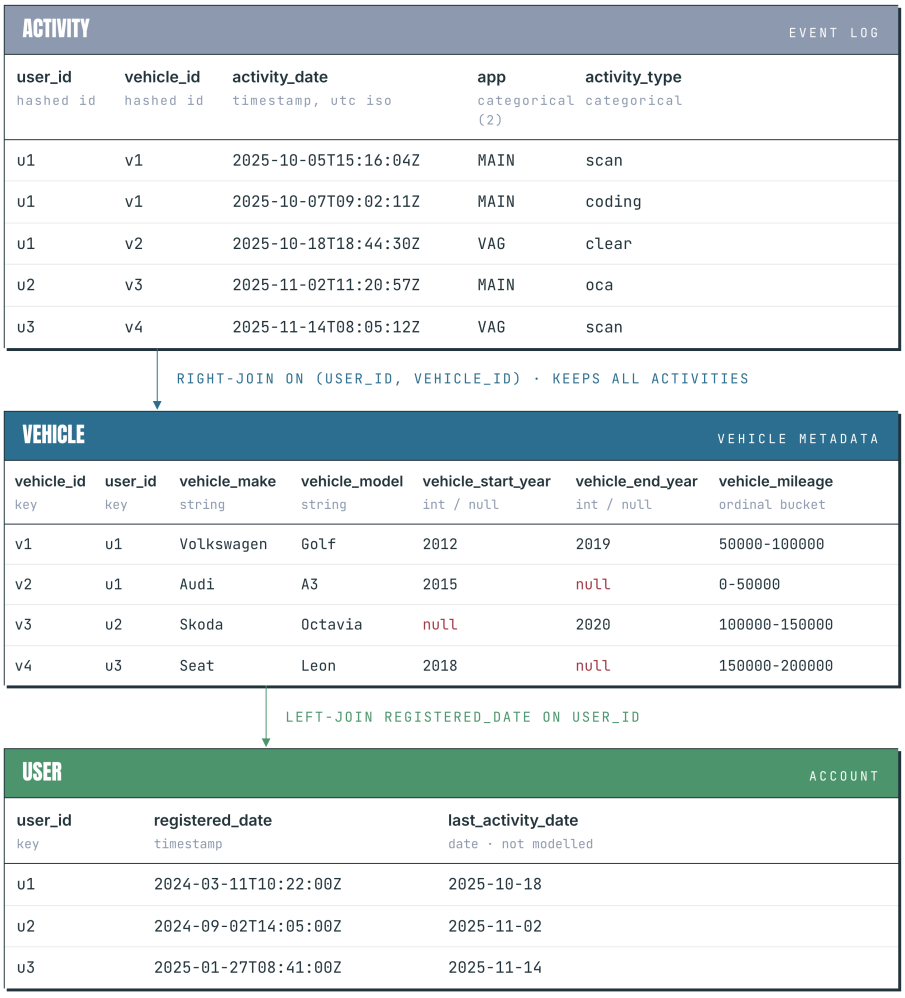


Figure 3.1: Merging three source tables into a single analysis table.

3.2 Separating personal from professional customers

The app serves both private owners and workshops, who use it differently and churn for different reasons. Only private owners, the personal segment, are modelled. The data hold no segment label, so one is built from how each customer spreads activity across vehicles. A private owner works on one or a few cars, while a workshop scans many. The rule has three steps (Figure 3.2).

1. *Collapse bursts.* A workshop often scans one car many times within minutes. Counted one by one, these scans make that car dominate the customer's activity. The workshop then looks like a private owner. A run of actions on the same vehicle, each less than 5 minutes after the previous one, therefore counts as one action. Thresholds of 10, 20 and 30 minutes gave the same segmentation. Bursts are collapsed only for segmentation, and the covariates use all actions.

2. *Measure concentration.* For customer i, V_i is the number of distinct vehicles, and s_{iv} is the share of the customer's activity on vehicle v. The effective number of vehicles, $N_i = 1/\sum_v s_{iv}^2$, is the reciprocal of the Herfindahl–Hirschman index [42]. It counts vehicles weighted by use. A customer who splits activity evenly over four vehicles has $N_i = 4$, and one with 90% of activity on a single car has N_i close to 1 (Appendix B.1).

3. *Classify.* A customer is personal if any of three conditions holds (Appendix B.2).

   A: an effective number of vehicles of at most 6.

   B: at least 80% of activity on the four most-used vehicles.

   C: at most 11 vehicles, the 90th percentile of vehicle counts (Section 4.1).

   Condition C alone classifies nine customers in ten as personal, so A and B decide only for customers with more than 11 vehicles. Condition B catches a case that A misses. A customer who uses four cars equally and scans twenty more once each has N_i just above 6, yet 80% of the activity sits on four cars.

Combining the conditions with OR favours sensitivity for private owners, at the cost of possibly keeping some workshops in the personal segment.

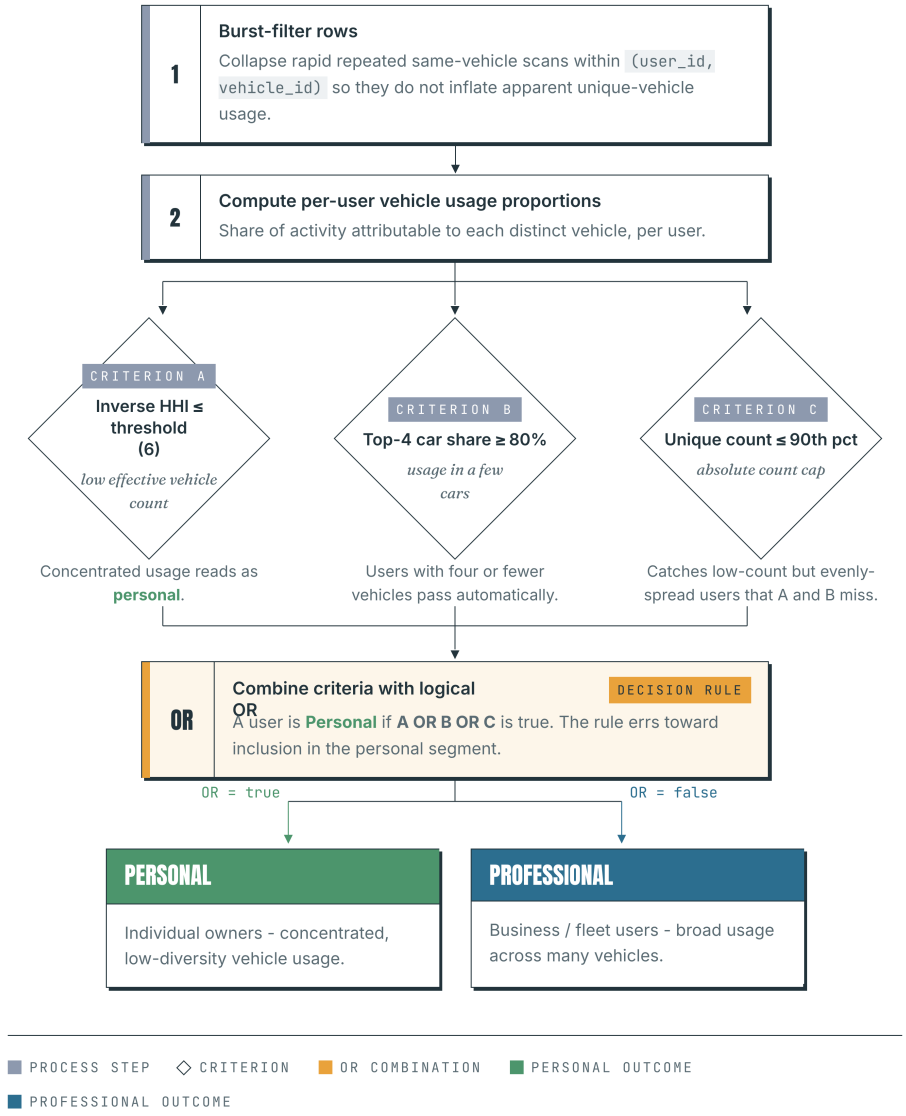


Figure 3.2: The segmentation procedure. Bursts are collapsed before concentration is measured, and the three conditions are combined with OR.

3.3 Defining churn

A customer is labelled as churned when a gap between active days reaches $\theta = 160$ days, or when their last activity is at least θ days before the cutoff, the last day in the data. The churn is then dated at the last active day before that spell, not at the day the 160 days run out. The 160 days are needed to confirm churn, but the modelled event is the start of the final inactivity, not the later date on which churn becomes confirmable. Dating churn at the confirmation date would add 160 days of follow-up after the customer had already left. The model would then learn to predict churn from inactivity that is part of the churn itself. Only the first churn is modelled, and activity after it is dropped.

The threshold sits between the 95th and 99th percentiles of the gaps between active days in the personal segment, 86 and 213 days (Section 4.2). Below the 95th percentile, ordinary returns would be labelled as churn. Above the 99th, the event would no longer be observable in the study window.

Seasonality decides the value within that range. Activity peaks in July and falls by about 41% to September (Section 4.2), so a customer active in summer can pause through autumn and return in winter without having left. A threshold of 160 days, about five months, lets such a pause pass, while 86 days would count it as churn. The exact value within the range is a judgement.

The same rule sets the end of follow-up. A customer whose last activity falls within θ days of the cutoff may still return, so whether they have churned is not yet known. Follow-up therefore ends at the cutoff minus θ, and customers still active then are censored at that date. Customers whose first activity comes after that date are excluded. They could not churn before follow-up ends, so keeping them would add survivors by construction and push the estimated survival towards one, the end-point bias described by Meža, Strle, and Meža [26] (Appendix B.3).

3.4 The person-interval frame

The modelling data have one row per customer and 14-day interval, from the first active date to churn or censoring (Figure 3.3). Each row holds the interval's start and end, the covariates and an event indicator for the following interval (Section 3.5). Survival data in this layout form a counting-process frame [4], which lets the covariates change from one interval to the next (Section 2.2.3).



Figure 3.3: The person-interval frame. Panel A cuts one customer's activity into 14-day rows. In Panel B, row 4, the first modelled row, covers days 42 to 56 but reads windows reaching back up to 56 days. Its covariates are computed on day 56, and its outcome is churn in days 56 to 70 (Table B.2).

Actions are aggregated into the interval they fall in, and several actions on one day count as one active day, called a session in the covariates. Intervals without activity are kept as rows of zeros, because inactivity itself signals churn. Splitting a customer into rows does not count the customer more than once, because each row covers a different stretch of time (Appendix B.4).

Covariates are rolling aggregates over the trailing 28- and 56-day windows that end at the end of the interval, so each row reflects a recent trend. Drift is the recent 28 days minus the prior 28 (Appendix B.5).

A narrower interval gives a finer time axis but more empty rows. The covariate windows stay fixed, so the width only changes how many rows each window spans. Row counts and ties were compared for 14- and 28-day widths (Section 4.3).

3.5 Restricting the sample

Four restrictions follow the segmentation: the removal of records with missing vehicle metadata, the inactivity rule and censoring date of Section 3.3, and the span filter below (Figure 3.4). Segmentation comes first, because a restriction applied before it could change which customers look personal.

The covariates summarise up to 56 days of past activity. For a customer seen for only a few days, these summaries are incomplete, and a model would learn that a short history predicts churn. The span filter therefore removes customers whose activity spans less than 56 days, and each customer is at risk in the model only from day 56. All results are conditional on staying at least 56 days. In survival terms this is delayed entry, or left truncation (Section 2.2.3).

Dropping activity records with missing vehicle metadata can also push a customer below 56 days even when their real activity runs longer. This is not fixed, since fixing it would mean imputing metadata for vehicles that were never identified.

Each row predicts the next interval, so the information a row uses always comes before the outcome it is scored on. A row whose windows close on day t uses activity up to day t, and its outcome is whether the customer churns in $(t, t + 14]$. The first modelled row covers days 42 to 56. Its windows close on day 56, the first day with a full history, and it predicts churn in days 56 to 70 (Table B.2). The frame therefore starts at day 42, the start of that row, although no covariate is computed and no outcome falls before day 56. Earlier rows

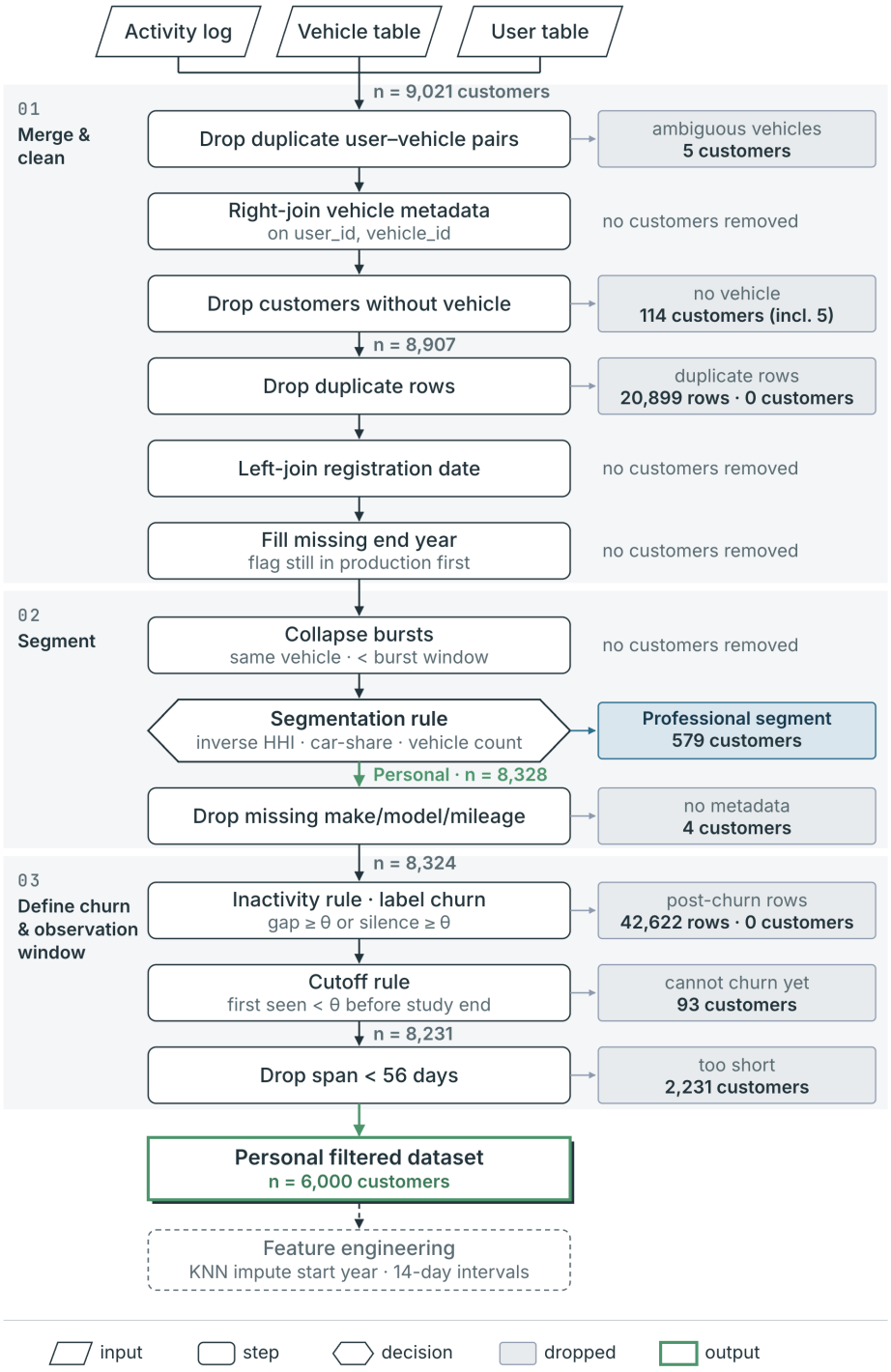


Figure 3.4: Sample construction. Grey boxes show what each step removes.

only fill the look-back windows. On the 28-day grid, the first row covers days 28 to 56 for the same reason.

During feature engineering, `vehicle_start_year` is imputed by k-nearest neighbours once per unique vehicle (encoding in Appendix C). The imputer and encoders are fitted before any split, so some information leaks into the imputed covariate. The leak does not reach the outcome, and it is the same in every fold. Refitting them inside each fold was judged too costly for a bias affecting one feature.

3.6 Covariates

The covariates fall into ten groups, listed with their windows and purposes in Figure 3.5. Four groups need a note.

Depth. Actions per session, which separates how much a customer does per visit from how often they visit.

Regularity. The mean, standard deviation and coefficient of variation of gaps between active days over 56 days. A row with fewer than two active days gets zeros and an indicator that they mean *not measurable*, since otherwise it would look like a row where every gap was identical.

Activity mix. Activity types are collapsed into six 56-day shares and a pooled *other* reference (Figure 3.6).

Vehicle portfolio. Each vehicle enters once, keyed on its first appearance within a customer, counting only vehicles seen up to the end of the row. Make enters as a share of distinct vehicles rather than an indicator, so a customer whose vehicles are 90% one make stays distinct from one at 55%.

Pairwise correlations are inspected, and variance inflation factors are checked against a cutoff of 5, the stricter of two common thresholds [7]. A variance inflation factor measures how much a coefficient's variance grows because its covariate can be predicted from the others. The make shares exceeded the cutoff (Section 4.4.2). The shares sum to one, so one make has to serve as the reference. Dropping a single make is not enough, because a few makes cover most personal vehicles. The three most common makes, Volkswagen, Audi and Škoda, are therefore kept as separate shares, and the rest are pooled into the reference (Figure 3.7).

FEATURE	WHAT IT MEASURES	WHY IT EXISTS
VOLUME		
n_sessions_0_28	Distinct active days in the recent 28d	Core engagement
sessions_drift_0_28_vs_28_56	Recent 28d minus prior 28d	Is the user speeding up or fading
DEPTH		
actions_per_session_0_28	Mean actions per active day	Separates depth from frequency
actions_per_session_drift_0_28_vs_28_56	The same drift, for depth	Deepening or thinning sessions
REGULARITY		
mean_gap_0_56 sd_gap_0_56 CV_gap_0_56	Typical, absolute and relative irregularity of gaps between active days	Erratic use precedes leaving
active_flag_0_56	At least 2 active days, so a gap exists	Distinguishes “no gap data” from “zero variability”
RECENCY		
recency	Days from last activity to interval end	How long since they last showed up
ACCOUNT		
account_tenure	Registered date to interval end	Long-standing vs new accounts
onboarding_delay	Register → first open	Instant adopter vs staller
TIMING		
weekend_share_0_28	Share of recent actions on Sat / Sun	Hobbyist vs weekday user
CONCENTRATION		
usage_concentration	Share of this interval's actions on the single most-used vehicle	One car vs the whole garage
ACTIVITY MIX · 56D SHARES		
prop_scan_0_56 prop_coding_0_56 prop_oca_0_56 prop_clear_0_56 prop_history_screen_0_56 prop_live_data_0_56 prop_other_0_56	Share of 56d actions by activity type	What kind of work they do
APP		
prop_main_0_56 prop_vag_0_56 prop_main_drift_0_28_vs_28_56	Main-app and VAG-app shares, and main-app drift	Which app the user relies on
VEHICLE PORTFOLIO		
prop_make_volkswagen prop_make_audi prop_make_skoda prop_make_other_rare	Share of vehicles by make	Which brands they work on
vehicle_mean_age vehicle_age_sd	Mean and spread of distinct-vehicle ages	Old fleet, mixed fleet
vehicle_mileage_ordinal	Mean of mileage-bucket lower edges, not exact km	Mileage arrives bucketed
prop_in_prod	Share of vehicles still in production	Newer fleet, longer support horizon
n_new_vehicles_0_28 days_since_last_new_vehicle	Vehicles first seen in the recent 28d, and time since the last one	Garage growth, not garage size

Figure 3.5: The covariates, the window each one reads, and the behavioural axis each is meant to capture.

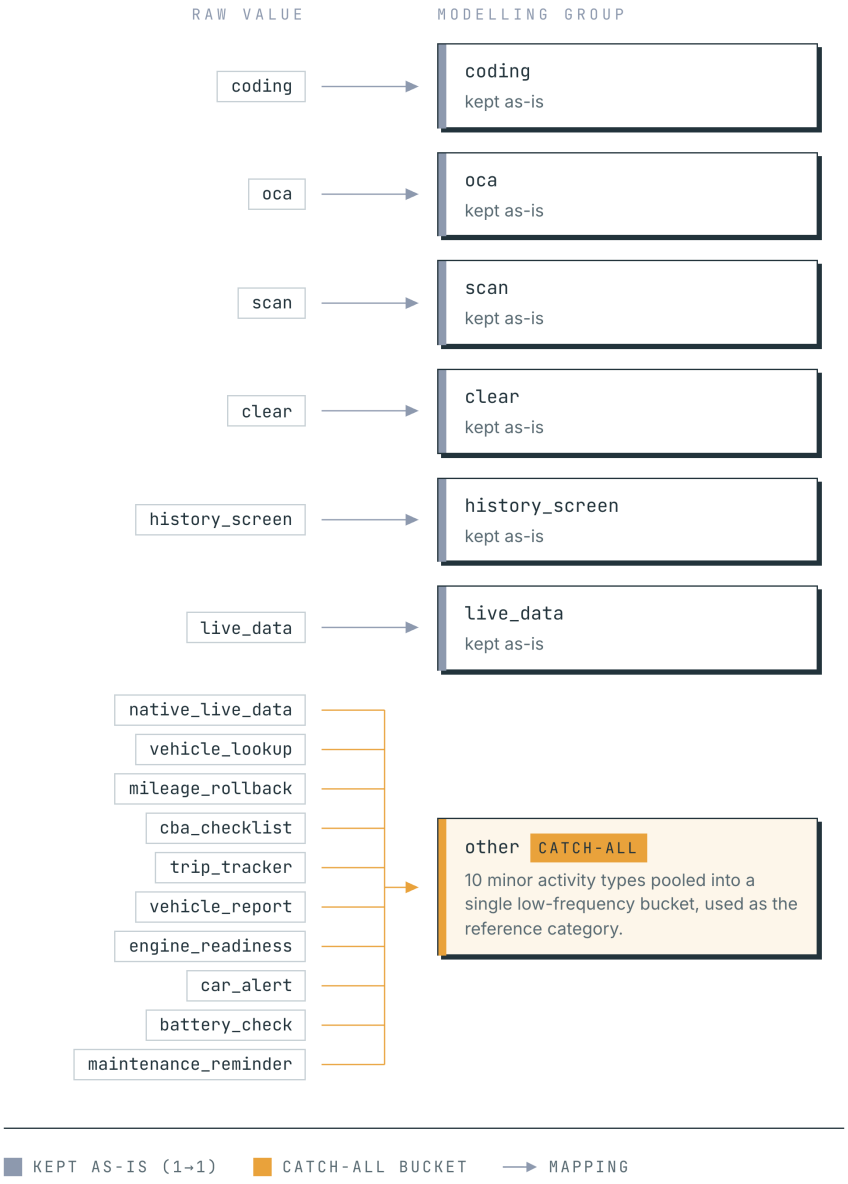


Figure 3.6: Activity type grouping. Six groups enter the model as proportions, and the remaining ten types are pooled into the omitted reference.

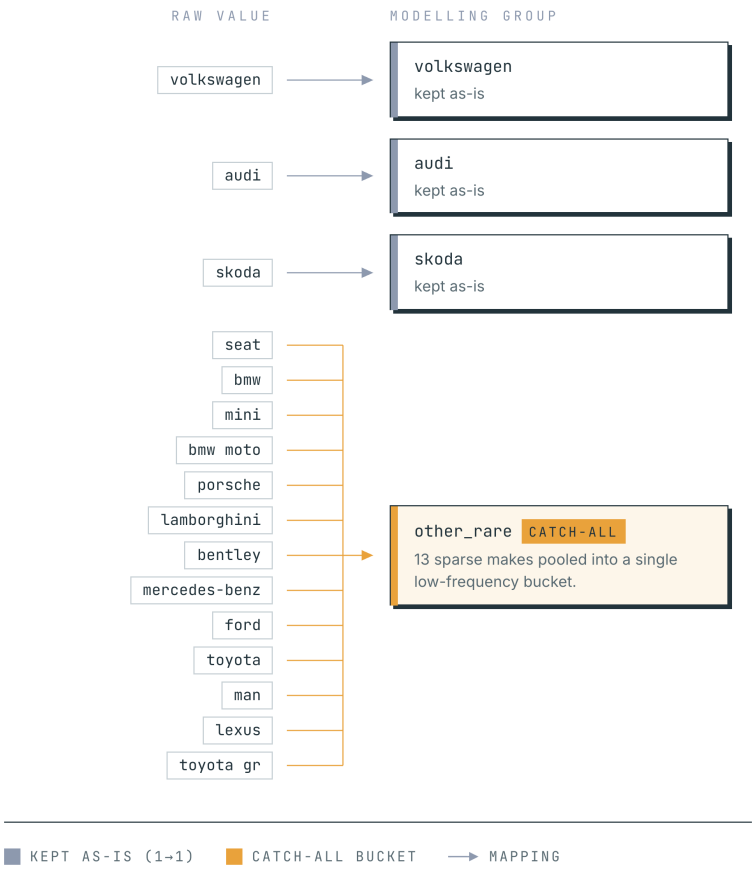


Figure 3.7: Vehicle make grouping. Three high-volume makes are kept separately, and the rest are pooled.

Of the 33 covariates built, 27 enter the model. Three are kept as reference categories (Table C.1), and three are held out because each duplicates information in the others (Table 3.1).

Covariate	Reason for exclusion
`recency`	Days from last activity to the interval end. The gap covariates measure the same behaviour, since the mean gap averages the running gap that recency reports (Section 4.4.2).
`account_tenure`	Equals the interval end plus the onboarding delay, so it is an exact linear function of the time axis and the retained onboarding covariate.
`CV_gap_0_56`	Equals $\mathrm{SD}/\mathrm{mean}$. With all three in the model, a hazard ratio for SD would hold mean and CV fixed, which is impossible. CV is dropped, since SD and mean are easier to interpret.

Table 3.1: Covariates built during feature engineering and excluded before fitting.

3.7 Model specification and estimation

The primary model, M0, is the Cox model of Section 2.2.1 with every covariate entered linearly, fitted by partial likelihood on the interval rows. Its covariates are updated at every interval, but each coefficient stays the same throughout follow-up, so M0 assumes proportional hazards.

The coefficients, standard errors and assumption tests come from one fit of each model on the full sample. Predictive performance comes from refitting the models in 5-fold cross-validation and scoring them on the held-out customers. The exception is M3, which only measures the size of a departure from proportional hazards and is fitted on the full sample alone (Section 3.7.1). Test statistics are computed once, on the full sample, not per fold. A partial likelihood depends on the size of the risk set, and a held-out fold holds only a fifth of it, so likelihoods from a fold and from the full sample are not comparable (Appendix B.9).

Churn is recorded on the 14-day grid, so many customers share a churn time (Section 4.3). With such ties, the partial likelihood cannot tell the order among the churners (Section 2.2.1). Efron's approximation handles them, and it is more accurate than Breslow's as the share of ties grows [24, p. 266]. The grouped model M2a handles ties exactly and serves as a check.

A customer contributes many rows, but the rows need no correction for correlation. With one event per customer, each customer enters each risk set once in the Cox models [43, p. 4]. In the grouped models, the likelihood is that of one binary outcome per row, provided the covariate process carries no information on the model's parameters [45, pp. 53, 60–61].

Standard errors are still clustered by customer. M0 sees only part of each customer's past, through its 28- and 56-day windows, and the model-based variance is valid only if the rest of that past does not affect the hazard [18, p. 210].

Regularisation was considered but not used, because shrinking the coefficients would make them harder to interpret.

Data are prepared in Python and BigQuery. The Cox and grouped models and the `cox.zph` tests are run in R with `survival` 3.8.6, which uses the newer implementation of the test (Section 4.5.1). M4 is fitted in Python with `xgboost` 3.4.1. Cursor and Claude were used for debugging, brainstorming and language editing, and their suggestions were checked before use. The code is available at `https://github.com/PadTo/Survival_Analysis_Thesis`. The data belong to OBDeleven and cannot be shared.

3.7.1 The model ladder

The thesis fits six models (Table 3.2). Each is compared with a model that differs from it in one respect, so each difference has one cause.

Model	Specification	Splines	β
M0	Cox, all covariates linear	no	constant
M1	Cox, restricted cubic splines on continuous covariates	yes	constant
M2a	Grouped complementary log-log, free intercept per interval, all covariates linear	no	constant
M2b	Grouped complementary log-log, free intercept per interval, restricted cubic splines on continuous covariates	yes	constant
M3	M0 with time-transform terms on the covariates the residual test flags	no	time-varying
M4	Gradient-boosted trees, same rows and folds	—	—

Table 3.2: The model ladder. M0 against M1 and M2a against M2b test linearity, the second assumption of RQ1. M0 against M2a checks the tie handling and the link. The tie check is needed because event times fall on interval boundaries. M3 measures the size of any departure from the first assumption. M1, M2b and M4 answer RQ2, with M4 as the performance benchmark.

M1 relaxes linearity. It replaces the straight line of each continuous covariate with a natural (restricted) cubic spline, a smooth curve made of cubic pieces joined at knots, so the effect can bend. Splines are used rather than martingale residuals (Section 2.4), because the models that test linearity, M1 and M2b, can then also be scored for RQ2. Each spline has three degrees of freedom (`ns()` in R), with knots at the 33rd and 67th percentiles and at the extremes. The knots are fixed in advance, so they are not chosen after seeing the results. They are computed on the training fold, so no test information enters the basis. Beyond the outer knots the curve is a straight line, which keeps it stable where data are sparse and for test values outside the training range.

Coinciding knots make the design rank deficient. A covariate is therefore splined only if it takes more than ten distinct values and its 67th percentile exceeds its minimum in every training fold (Appendix B.7). The rest stay linear in M1 and M2b as well (Section 4.5.2).

M2a treats each interval row as a binary outcome, churn or no churn, fitted with a complementary log-log link and a separate intercept for each interval. It is the discrete-time version of the Cox model. It estimates the same β as M0, but it handles tied churn times exactly, because it never needs their order. M2b adds the splines, so M2a against M2b repeats M0 against M1 under a different link (Figure 3.8).

M3 lets the coefficient of each covariate with rank-transform $p < 0.1$ in the proportional hazards test change over follow-up, as $\beta + \beta^{\text{tt}}\, g(t)$ with $g(t) = t$, and with $g(t) = \log t$ as a check (Figure 3.9). The estimate of β^{tt} shows how far the effect drifts.

M4 is gradient-boosted trees (XGBoost), which treat each interval row as a yes/no outcome and make neither assumption. It is fitted on the same rows and folds as the other models, so any performance difference comes from the model. The probability of churn in a row is the hazard for that interval. Any learner that minimises log loss on the rows therefore fits the same likelihood as the grouped models (Appendix A.5.4).

3.7.2 Validation and diagnostics

Events per variable, the number of churn events per estimated coefficient, are checked against a floor of 10, below which Cox coefficients become unstable [29] (Section 4.4). M0, M1, M2a, M2b and M4 are scored out of fold, on five folds assigned by customer and stratified on event status, so no model is scored on a customer it was fitted on. M4's seven hyperparameters are tuned

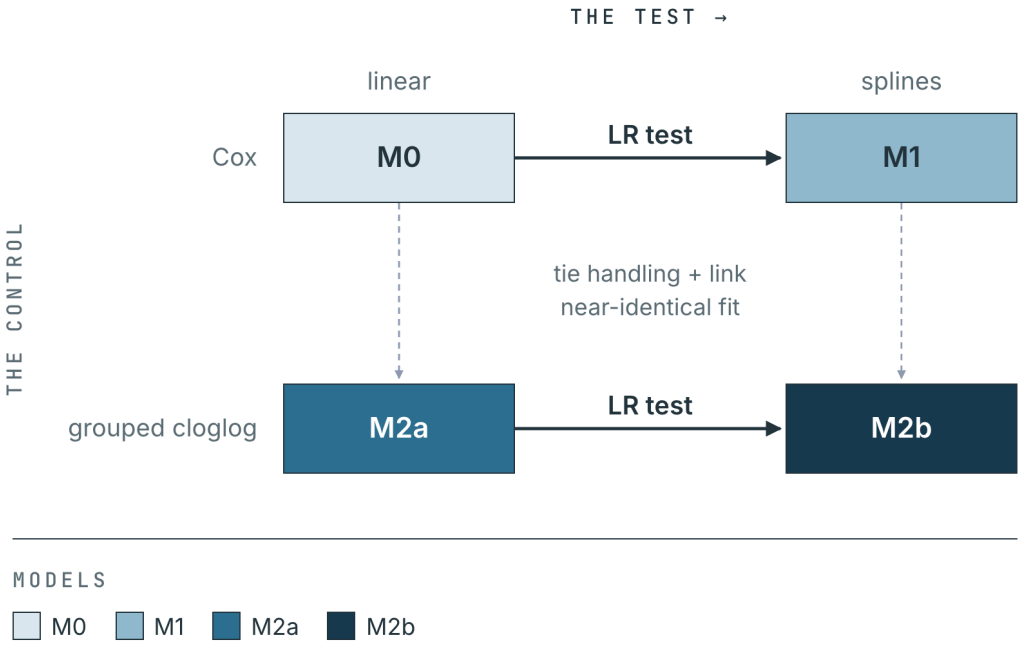


Figure 3.8: The four models behind the second assumption. Moving across a row relaxes linearity and is the test. Moving down a column changes the tie handling and the link, and serves only as a control.

on log loss in an inner five-fold split, also by customer, so tuning never sees the held-out fold either (Figure 3.10, Section 4.5.3).

M4 also receives the interval index, which stands in for the baseline the Cox models estimate, so that any deficit in M4 does not come from a missing time effect. The index is known at the start of each interval and leaks nothing. It also absorbs selection, since customers still at risk late tend to be the least likely to churn.

Each row is scored with the probability of churn in the interval it predicts. The grouped models give it directly. For the Cox models, the baseline is recovered with the Breslow estimate (Appendix B.8).

Three quantities are computed on each held-out fold and averaged over the five folds. Discrimination is the area under the ROC curve (AUC), the probability that a row with churn gets a higher predicted risk than a row without. Calibration is measured by the calibration slope, the slope of held-out outcomes on the linear predictor. A slope below one means the predictions are too spread out. Calibration is also plotted over 20 equal-count bins (Section 4.5.3). The Brier score, the mean squared error of the predicted probabilities, reflects both. At a low event rate most of it cannot be removed by

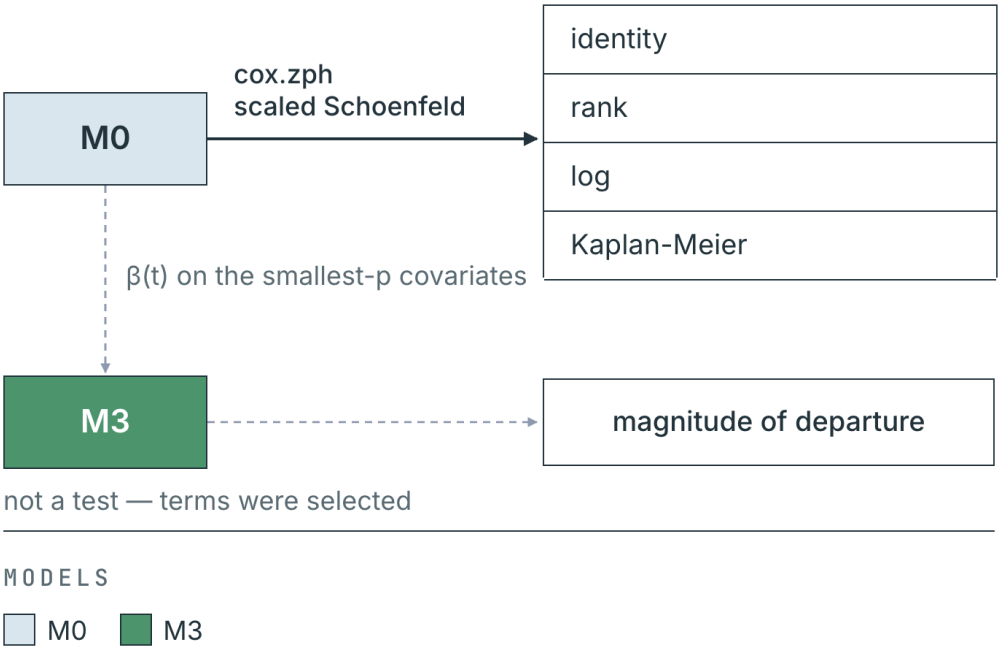


Figure 3.9: What is fitted and what is tested for the first assumption. The test runs on M0. M3 fits the non-proportional alternative and reports the size of any departure, and is not a second test, since its terms are chosen from M0's results.

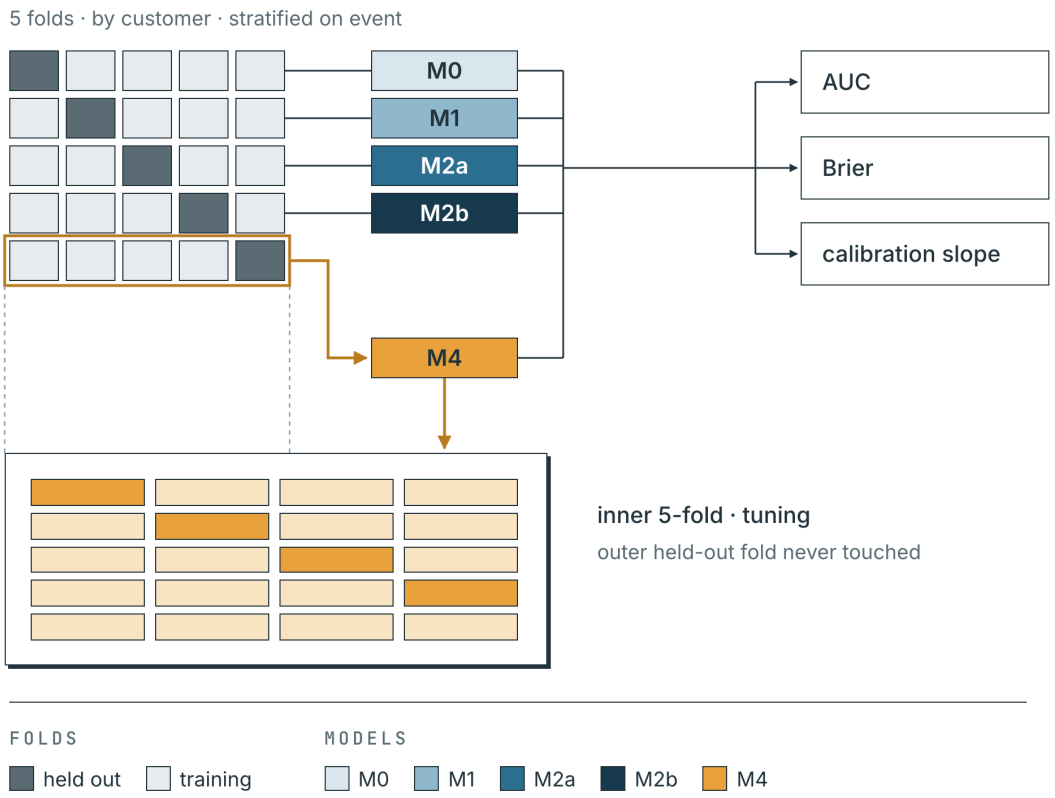


Figure 3.10: The cross-validation protocol. Folds are assigned by customer, and only M4 divides its training folds a second time to tune hyperparameters without touching the held-out fold.

any model, so it is reported but not used to separate models. M4 has no coefficients, so it is read through out-of-fold SHAP values [23], which split each prediction into the contribution of each covariate. Their ranking is compared with M0's coefficient ordering. Linearity is screened by likelihood ratio tests of M0 against M1 and M2a against M2b. Whether curvature is kept is decided by out-of-fold AUC, Brier score and calibration.

Proportional hazards is tested on M0 with scaled Schoenfeld residuals [11]. At each churn time, a covariate's Schoenfeld residual is the churner's value minus the average over everyone still at risk, weighted by their fitted risk. If the covariate's effect is constant, the scaled residuals show no trend over time. The test regresses them on a transform of time, per covariate and for all covariates jointly, and the choice of transform can change the conclusion. Rank does best under heavy censoring [28], as in these data (Section 4.1), so it is primary among the four transforms computed. Per-covariate results are reported even when the global test is null, since one drifting covariate can hide among many stable ones [28]. A null may mean the assumption holds or that the test, whose power is often low [5], could not detect a departure, so it is reported with the departure M3 fits.

3.8 Ethics

OBDeleven provided the data under a mutual non-disclosure agreement, attached as an appendix to Stockholm University's collaboration agreement for degree projects, on condition that they stay confidential [8, p. 220]. Customers are identified only by pseudonymous IDs, no names or contact details were shared, and results are reported only in aggregate, so no individual can be identified [8, pp. 332, 345].

The IDs are pseudonymous, not anonymous, since OBDeleven can link them to customers, so the data remain personal data under the GDPR [10, Recital 26].

4

Results

Chapter 3 asserted a segmentation rule, a churn threshold, an interval width and a covariate set. This chapter provides evidence for each in turn, then presents the fitted models in Section 4.5.

4.1 The estimation sample

	Personal	Professional
After segmentation, Eq. (B.5)	8,328	579
After the inactivity rule, censoring and entry filter	6,000	514
Removed	2,328	65
Events	2,774	254
Censored	3,226	260
Customers with an event	46.2%	49.4%
Person-interval rows	124,783	—
Rows carrying an event	2.22%	—

Table 4.1: Customers before and after the jointly applied restrictions of Section 3.5. The professional segment is not modelled.

The restrictions of Section 3.5 – the missing-metadata drop, the inactivity rule, the cutoff-minus-θ censoring and the 56-day entry filter – cost the personal segment 2,328 of its 8,328 customers, or 28.0% (Table 4.1). The remaining 6,000 contribute 124,783 person-interval rows. Just under half of them churn, and 2.22% of rows carry the event.

Every `vehicle_id` in the activity log matches a row in the vehicle table, so the right join of Section 3.1 loses no activity. It cannot reach the 7,762 activity rows that carry no `vehicle_id` at all. `vehicle_end_year` is missing for 7.31% of vehicles, which is the share treated as still in production. Five customers had a single `vehicle_id` with conflicting metadata, and all their data was dropped.

Condition C of (B.5) needs a cut-off q on distinct vehicles per customer. The 90th percentile of that count sets it at 11 and the 95th percentile is 20.

The two segments separate clearly (Figure 4.1). No personal customer exceeds an effective count of 10, while professional customers reach 226.

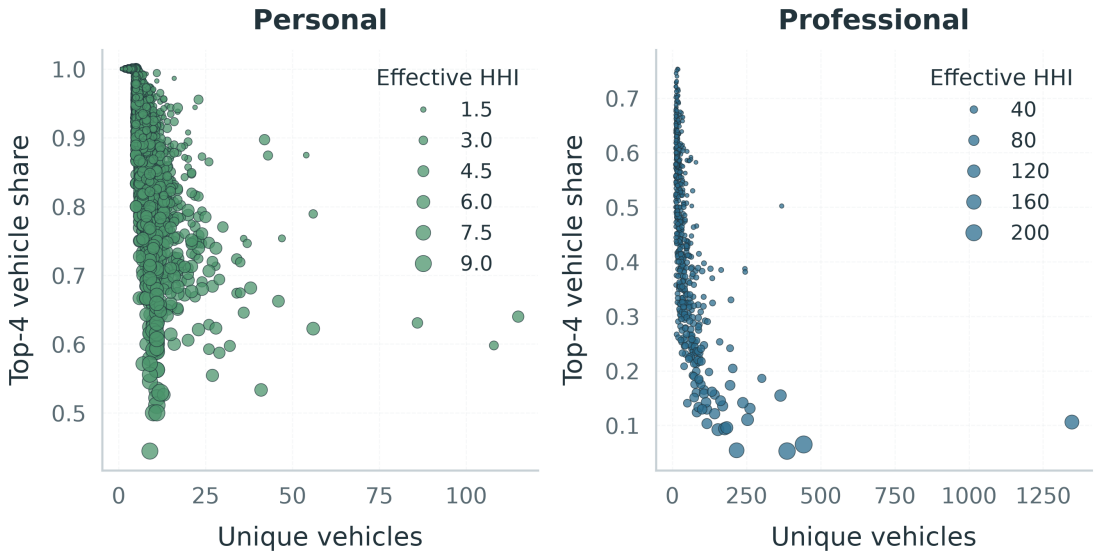


Figure 4.1: Customers by distinct vehicle count and top-four vehicle share, sized by effective vehicle count. The horizontal scales differ.

The segments also use the app differently, although the rule never sees the activity types. Scans make up 49.4% of personal actions but 40.5% of professional ones, while one-click apps rise from 9.2% to 24.6% and coding from 7.8% to 14.2% (Figure 4.2). The six modelled activity groups cover 93.1% of personal actions.

The professional segment appears in Table 4.1 for comparison only. Its 254 events give about 203 per training fold, or 7.5 per covariate, below the floor of 10 (Section 4.4).

4.2 How long an absence has to be for churn

Half of all personal returns come within 4 days and three-quarters within 15, yet the mean gap is 18.4 days, more than four times the median, and the 99th percentile reaches 213 (Table 4.2, Figure 4.3). The 160-day threshold falls between the 95th percentile (86 days) and the 99th (213).

The margin above the 95th percentile is seasonal. Personal activity peaks in July at 53,845 actions and bottoms out in September at 31,647, a drop of about 41%. The professional segment follows a similar trend, from a July peak of 10,276 to an August low of 6,826 (Figure 4.4). A customer active at the summer peak who pauses through autumn and returns in winter produces a gap well

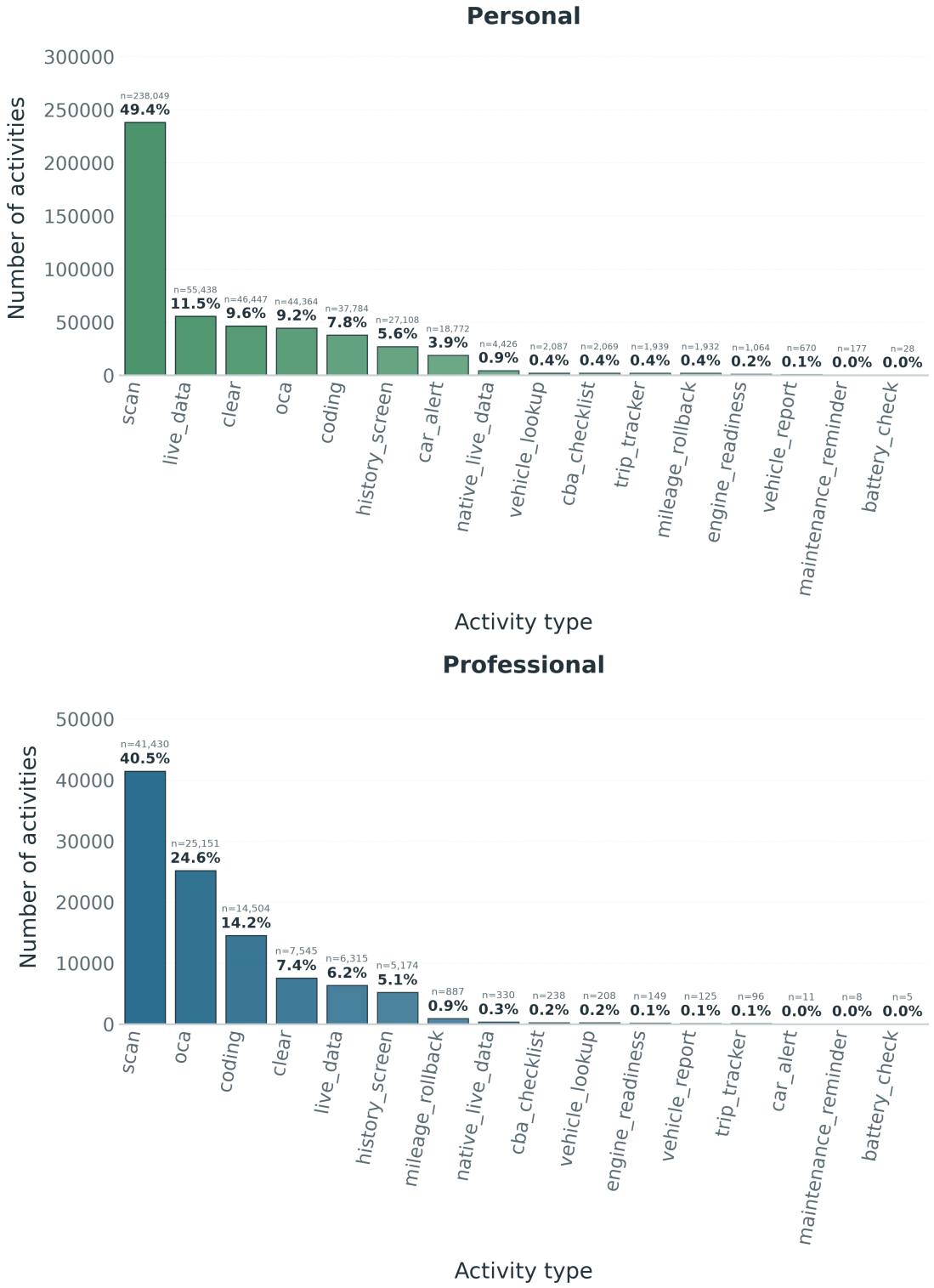


Figure 4.2: Activity types by segment, with counts and shares. Personal above, professional below. The vertical scales differ.

Statistic	Personal	Professional
Count	172,331	35,324
Mean	18.4	8.5
SD	41.3	17.4
Minimum	1	1
25th percentile	1	1
Median	4	3
75th percentile	15	8
90th percentile	47	20
95th percentile	86	33
99th percentile	213	83
Maximum	937	425

Table 4.2: Days between consecutive active days by segment, before churn filtering.

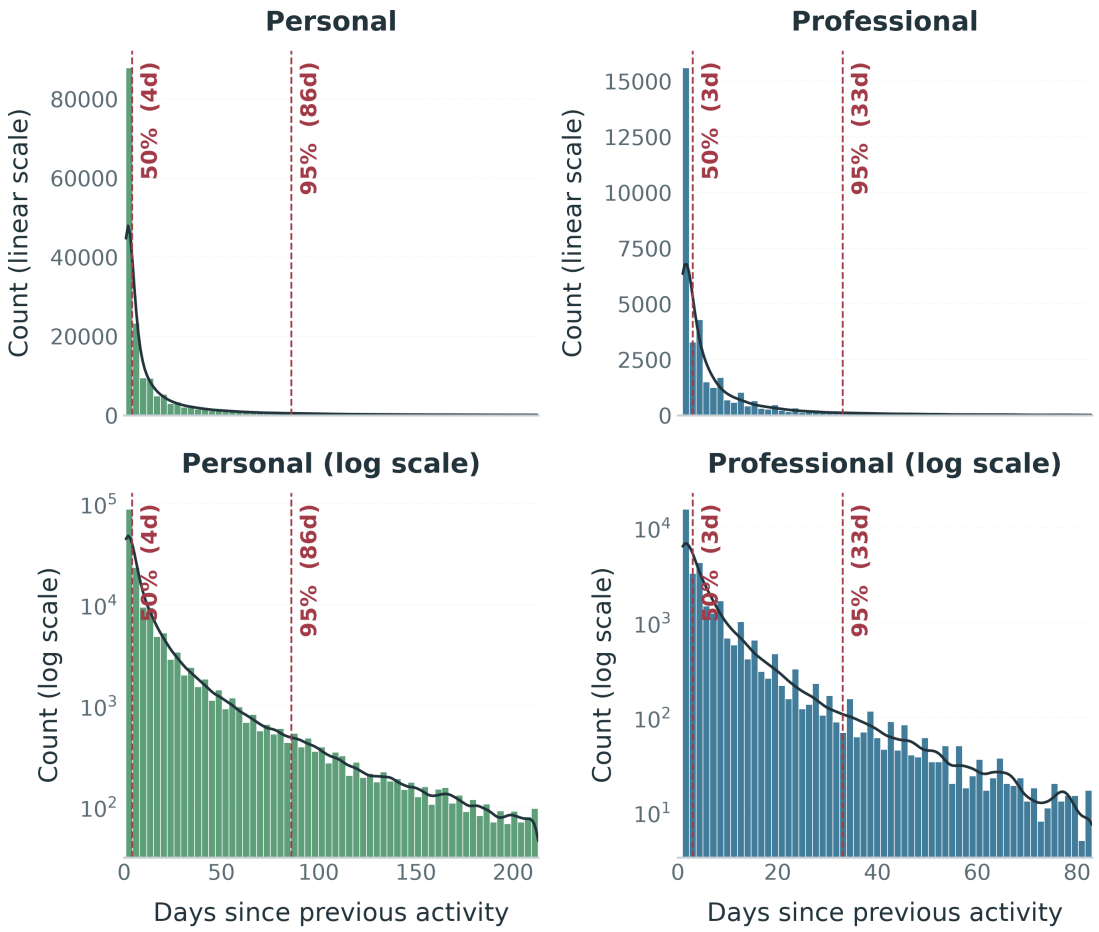


Figure 4.3: Gaps between consecutive active days on linear (above) and log (below) scales, with median and 95th-percentile lines, trimmed at the 99th percentile.

over 100 days without having left. A threshold of 86 days would record that customer as churned, while 160 days would not.



Figure 4.4: Actions by calendar month. Personal on the left, professional on the right, with the peak month highlighted.

The professional segment returns faster on every measure, with a median gap of 3 days against 4 and a 95th percentile of 33 against 86, so a single threshold fitted to the combined distribution would be too wide for professional customers and too narrow for personal ones. Segmentation has to precede churn thresholding.

The gap covariates are computed on the filtered data within a trailing 56-day window, so Table 4.3 is the distribution they are built on, with a median of 4 days and a 90th percentile of 39.

4.3 Cutting time into intervals

Halving the interval width nearly doubles the rows, from 63,927 to 124,783 (Table 4.4). Fourteen days was the natural first choice, since it equals the 75th percentile of gaps between active days, so three returns in four come within 14 days (Table 4.3). It also divides the rolling windows evenly, giving the 28-day window exactly two rows and the 56-day window four.

Events can only fall on interval boundaries, so wider intervals mean more ties. At 14 days the 2,774 events fall on 51 distinct times, 54 per time on average and up to 210 (Figure 4.5), and doubling the width roughly doubles these counts.

Statistic	Gap (days)
Count	150,522
Mean	13.7
SD	23.8
Minimum	1
25th percentile	1
Median	4
75th percentile	14
90th percentile	39
95th percentile	65
99th percentile	123
Maximum	160

Table 4.3: Days between consecutive active days, personal segment, after churn filtering.

Width (days)	Rows	Mean intervals	Median	p25	p75
14	124,783	20.8	18	9	30
28	63,927	10.7	9	5	15

Table 4.4: Intervals per customer by width, after the entry filter. Both widths carry the same 6,000 customers and 2,774 events.

Neither width comes close to distinct event times, which is why M2a is fitted as a check on Efron's approximation (Section 3.7).

4.4 Constraints on the modelling frame

What can be fitted on the frame depends on how much of it is inactivity, how strongly the covariates correlate, and whether the events support twenty-seven covariates. Roughly 2,219 of the 2,774 events fall in each 80% training partition, an events-per-variable (EPV) ratio near 82 at 27 covariates, far above the floor of 10 recommended by Peduzzi et al. [29], and still about 35 once the 36 spline parameters of M1 and M2b are added.

4.4.1 Empty activity windows

The volume and timing covariates are built from 28- and 56-day windows, and Table 4.5 shows how often those windows are empty.

The customer did nothing in the last 28 days on 48,981 rows and had fewer than two active days in the last 56 on 49,227, about two rows in five. On those rows, the volume and timing covariates are zero or flagged as not measurable,

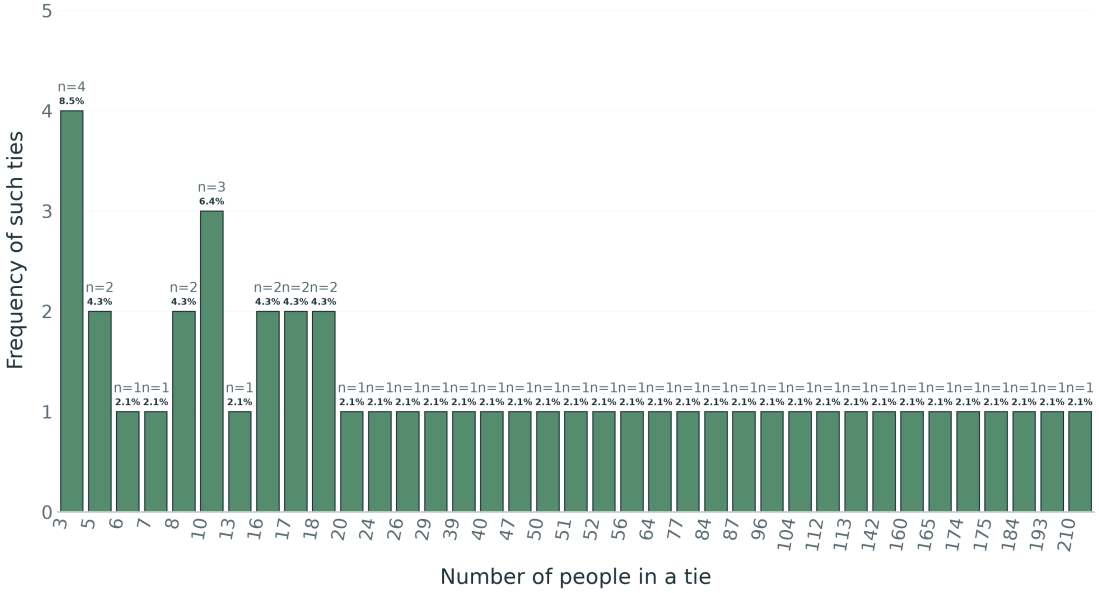


Figure 4.5: Number of customers sharing each event time. The four untied events are omitted.

	Rows	Customers
Rolling windows		
No actions in the last 28 days	48,981 (39.3%)	
Fewer than two active days in the last 56 days	49,227 (39.5%)	
No actions in the 28 days before that (drift)	45,326 (36.3%)	5,259 (87.7%)
Empty 14-day intervals		
At least one empty interval		5,841 (97.4%)
Median share of a customer's rows that are empty		66.7%
Empty-run length, median / p90 / max	2 / 6 / 11	
Empty runs of a single interval	38.5%	

Table 4.5: Rows and customers with empty activity windows, out of 124,783 rows and 6,000 customers. Windows are counted back from the end of each row, and an empty interval is a 14-day row with no actions.

so across much of the frame they separate inactive from active customers but say little about how the inactive ones differ.

Drift is the recent 28-day count minus the prior one, so it equals the recent count whenever the prior window is empty, which happens on 45,326 rows (36.3%) and for 5,259 customers (87.7%). The windows use each customer's full history, and no prior window reaches back before the first activity (Appendix B.6), so these rows reflect real inactivity rather than a short history.

4.4.2 Collinearity

Table 4.6 separates the two sources of collinearity by changing the make grouping and the treatment of raw action counts one at a time.

With six vehicle makes kept separately, Volkswagen and Audi stood at 19.1 and 18.3, and Škoda at 7.5. Pooling SEAT, the BMW group and Premium into one low-frequency category brings them to 2.69, 2.70 and 1.67. Six make proportions sum to one and two of them dominate, so no choice of reference category could have broken that dependence.

The volume block mattered less. With the raw action count, its drift and the actions-per-session ratio all present, sessions and the ratio sat at 5.1 and 5.0. Dropping the raw count and its drift takes them to 3.1 and 3.3.

Only recency (6.4) and gap CV (5.7) stay above 5, and both are excluded (Section 3.6). Every covariate entering the model is below 5, with the highest being gap SD at 3.9. No pairwise correlation among the 26 covariates in Figure 4.6 exceeds 0.64 in absolute value, so the inflation comes from variance shared across several covariates rather than from any one pair.

4.5 Model results

RQ1 asks whether the two assumptions given in Section 2.4 hold. RQ2 asks what relaxing them buys on customers the models have not seen, using the protocol of Section 3.7.2. Table 4.7 summarises the results.

4.5.1 Assumption 1: does the hazard ratio stay constant?

The Schoenfeld test runs on M0 alone (Figure 3.9), under four time transforms in `survival::cox.zph`, since the transform can change the conclusion [28].

No transform rejects the null, with statistics between 22.6 and 28.7 against a critical value of 40.1 (Table 4.8).

Covariate	Six makes raw counts (35)	Six makes ratio (33)	Pooled raw counts (32)	Pooled ratio (30)
Recency	6.51	6.45	6.51	6.44
Gap CV (56d)	5.74	5.65	5.74	5.65
Gap SD (56d)	3.92	3.91	3.92	3.91
Scan proportion	3.86	3.79	3.86	3.78
Active flag (56d)	3.50	3.49	3.50	3.49
Actions per session (28d)	**5.00**	**3.29**	5.00	3.29
Sessions (28d)	**5.05**	**3.07**	5.05	3.07
Actions-per-session drift	**4.21**	**2.99**	4.21	2.99
Audi proportion	**18.33**	18.32	**2.70**	2.70
Volkswagen proportion	**19.08**	19.08	**2.69**	2.69
Usage concentration	2.31	2.29	2.31	2.29
Gap mean (56d)	2.28	2.27	2.28	2.27
Session drift	**3.36**	**2.05**	3.36	2.05
Account tenure	1.95	1.95	1.95	1.95
One-click app proportion	1.97	1.93	1.97	1.93
Days since last new vehicle	1.88	1.88	1.87	1.87
Live-data proportion	1.83	1.81	1.83	1.80
Škoda proportion	**7.52**	7.52	**1.67**	1.67
Mean vehicle age	1.62	1.62	1.58	1.58
New vehicles (28d)	1.56	1.56	1.56	1.56
Coding proportion	1.54	1.53	1.54	1.53
Vehicle age SD	1.47	1.47	1.47	1.47
Clear proportion	1.48	1.46	1.48	1.46
History-screen proportion	1.45	1.44	1.45	1.44
Main-app drift	1.44	1.38	1.44	1.38
Main-app proportion	1.32	1.32	1.29	1.29
Mean mileage ordinal	1.52	1.52	1.29	1.29
Onboarding delay	1.20	1.20	1.20	1.20
Weekend share (28d)	1.18	1.17	1.18	1.17
In-production proportion	1.17	1.17	1.16	1.16
SEAT proportion	**5.29**	5.29	**pooled**	pooled
BMW group proportion	**4.03**	4.03	**pooled**	pooled
Premium proportion	**1.26**	1.26	**pooled**	pooled
Actions (28d)	**5.05**	**replaced**	5.05	replaced
Action drift	**4.17**	**replaced**	4.17	replaced

Table 4.6: Variance inflation factors under four covariate sets (covariate count in brackets). Makes are kept separate or pooled, and the raw action count and its drift are kept or replaced by actions per session. The last column is the fitted set, reduced to 27 in Section 3.6. Black bold marks the make regrouping (columns 1 and 3), red bold the switch to the ratio (columns 1 and 2).

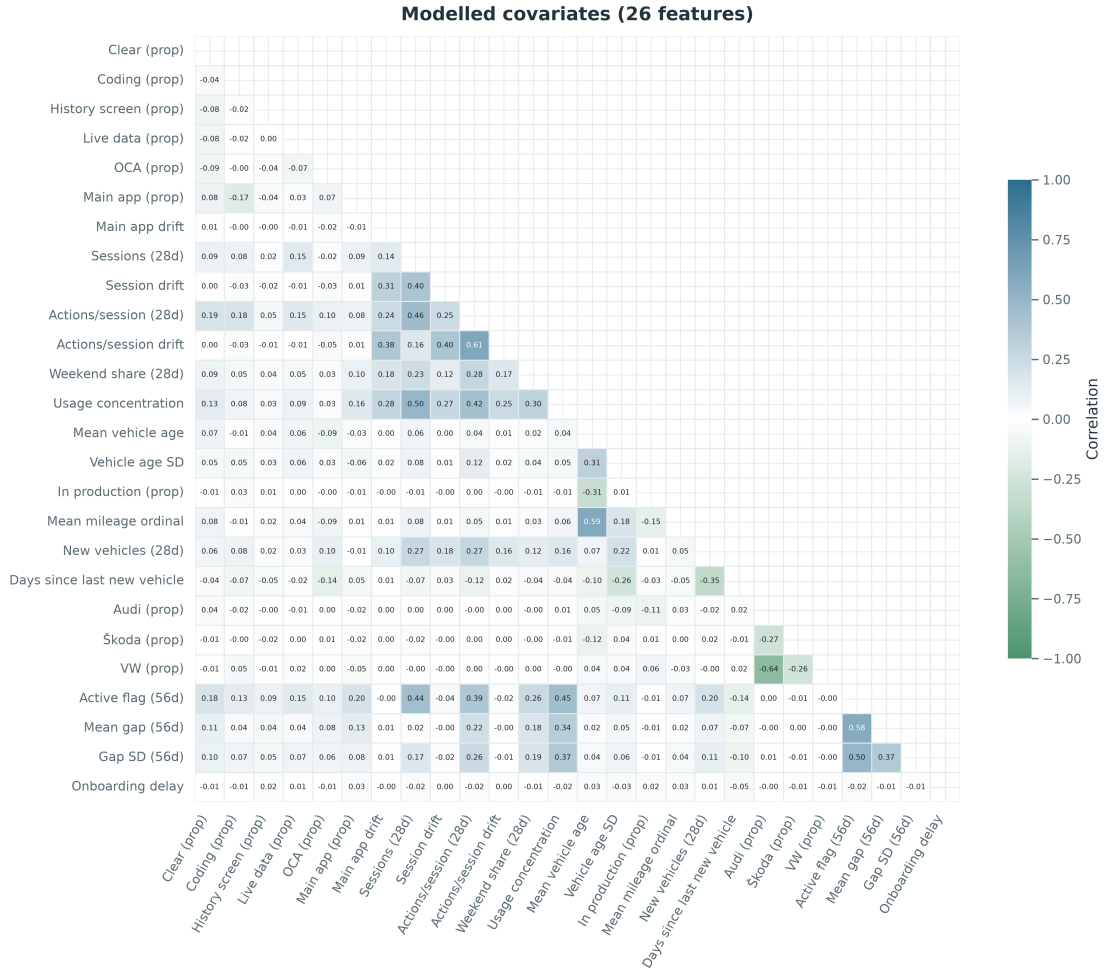


Figure 4.6: Pairwise Pearson correlations across 26 of the 27 modelled covariates, without the scan proportion. No pair exceeds 0.64 in absolute value.

Comparison	Result	Verdict
RQ1 – PROPORTIONAL HAZARDS		
Schoenfeld global, four transforms	χ^2_{27} from 22.6 to 28.7, p from 0.705 to 0.374	Not rejected
M3, largest drifting covariate	HR per SD 0.89 $\rightarrow$ 1.11 (identity)	Possible reversal
RQ1 – LINEARITY		
M0 against M1, Cox link	$\chi^2_{36} = 109.8$, $p = 2.1 \times 10^{-9}$	Rejected
M2a against M2b, cloglog link	$\chi^2_{36} = 111.6$, $p = 1.1 \times 10^{-9}$	Rejected
M0 against M2a, coefficients	$r = 0.99999$	Ties not the cause
RQ2 – WHAT RELAXING THEM BUYS		
M1 against M0, out of fold	$+0.0019$ AUC, better in 4 of 5 folds	Negligible
M4 against M0, out of fold	$+0.0053$ AUC, better in 5 of 5 folds	Small gain

Table 4.7: Summary of the tests, detailed in Sections 4.5.1 to 4.5.3.

Transform	χ^2	df	p
Identity	22.63	27	0.705
Rank	28.00	27	0.411
Log	28.21	27	0.400
Kaplan–Meier	28.73	27	0.374

Table 4.8: Global test of proportional hazards on M0. The null expectation is 27 and the 5% critical value 40.1.

The global test can hide one drifting covariate among many that do not, a pattern Park and Hendry [28] find in a fifth of the analyses they replicate, so covariate-specific results are also reported.

Covariate	Identity	Rank	Log	KM
One-click app proportion	**0.043**	**0.045**	0.051	0.053
Gap SD (56d)	0.211	0.094	0.084	0.075
Volkswagen proportion	0.118	0.099	0.116	0.108
Scan proportion	0.384	0.127	0.069	0.072

Table 4.9: The four smallest covariate-specific p-values, ordered by the rank transform. Bold marks $p < 0.05$. The other 23 do not reach 0.1 under any transform (all 27 in Figure 4.7).

Four covariates have $p < 0.1$ under at least one transform (Table 4.9). Only the one-click app proportion falls below 0.05, under the identity and rank transforms but not the log or Kaplan–Meier. Three of the four are below 0.1 under the rank transform, the primary one, and these are carried to M3.

With 27 covariates under four transforms, one p-value near 0.04 is about what chance alone would give. The test uses `cox.zph` from `survival` 3.8.6, whose newer implementation over-rejects when covariates are correlated [25], as these are (Section 4.4.2), while low power [5] works the other way. The one-click result is therefore a possible violation, not a confirmed one.

M3 measures the size of each departure. It refits M0 with a time-transform term on those three covariates, so a constant β_k becomes $\beta_k(t) = \beta_k + \beta_k^{\mathrm{tt}}\, g(t)$, and reports how far that coefficient moves between the first and last event time. This is not a hypothesis test, since the covariates were chosen from M0's results.

The hazard ratios for the Volkswagen proportion and gap SD start at about 1 and fall below it (Table 4.10). The one-click app coefficient travels $+1.20$, and the hazard ratio per unit runs from 0.52 at the first event time to 1.74 at the last, crossing 1 at about day 455. It lowers the hazard early in follow-up and may raise it late. The log transform gives the same pattern at a smaller magnitude, crossing 1 at about day 574, so it is not an artefact of the time scale.

These hazard ratios are per unit of x, which for a proportion is its whole range from 0 to 1, while its standard deviation is only 0.18. Per standard deviation under the identity transform, the hazard ratio moves from 0.890 (95% cluster-robust CI 0.841–0.941) at the first event time to 1.105 (0.930–1.313) at the last. Rescaling shrinks the effect but keeps its sign, so the hazard ratio still crosses

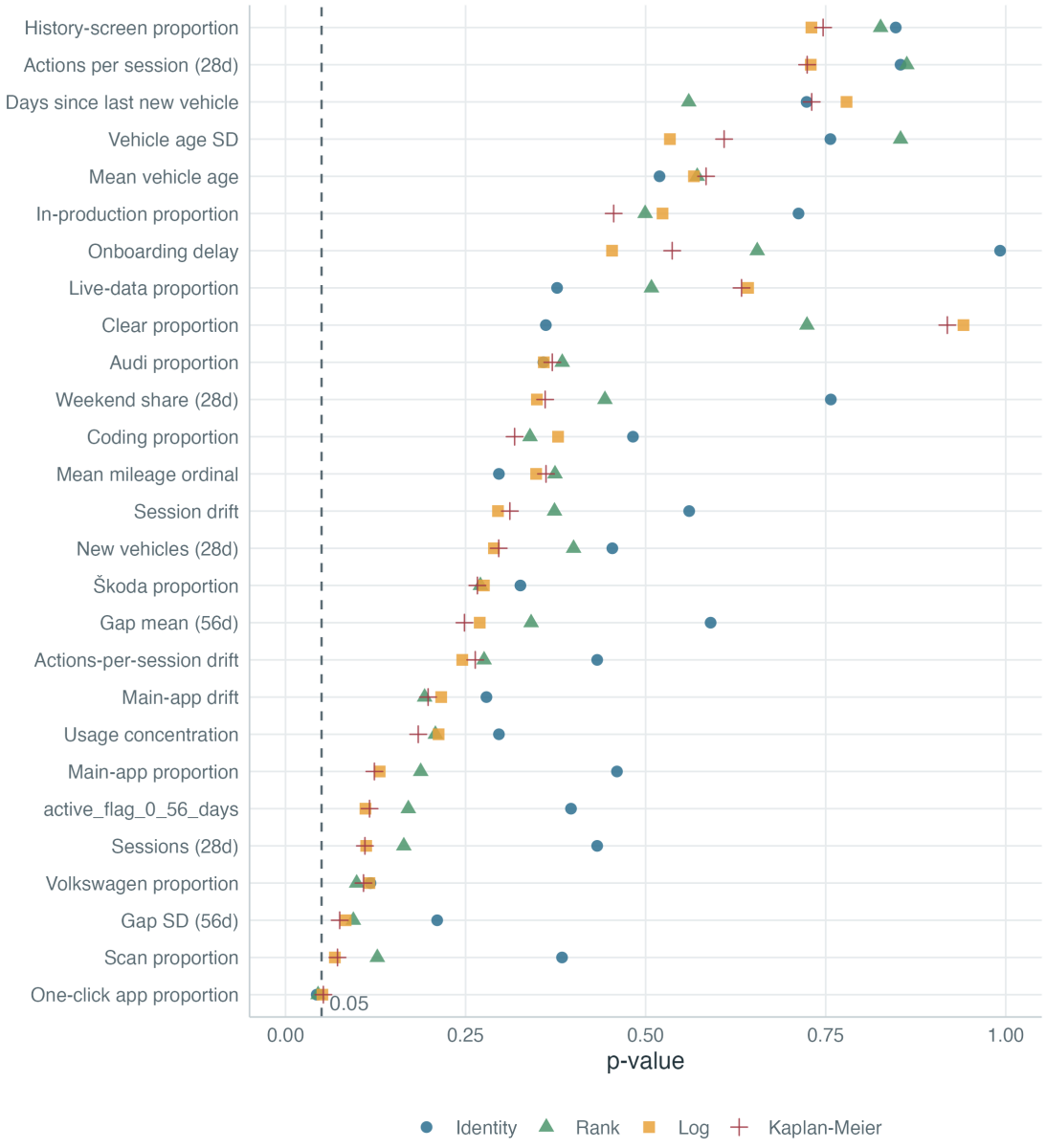


Figure 4.7: Covariate-specific p-values under each transform. The dashed line is 0.05.

Covariate	$\hat{\beta}$ (M3)	$\hat{\beta}^{\mathrm{tt}}$	p	$\Delta\log\mathrm{HR}$	HR start $\to$ end
Identity transform, $g(t) = t$					
One-click app proportion	-0.7392	$+1.62 \times 10^{-3}$	**0.038**	$+1.204$	$0.52 \to 1.74$
Volkswagen proportion	0.0543	-5.99×10^{-4}	**0.119**	-0.444	$1.02 \to 0.65$
Gap SD (56d)	-0.0018	-4.36×10^{-5}	**0.223**	-0.032	$1.00 \to 0.96$
Log transform, $g(t) = \log t$					
One-click app proportion	-2.0893	$+0.329$	**0.048**	$+0.874$	$0.47 \to 1.11$
Volkswagen proportion	0.5387	-0.118	**0.116**	-0.315	$1.06 \to 0.78$
Gap SD (56d)	0.0469	-0.0114	**0.081**	-0.030	$1.00 \to 0.97$

Table 4.10: M3 on the three covariates with rank-transform $p < 0.1$. $\Delta\log\mathrm{HR}$ is the change in log hazard ratio between the first and last event time (days 56 and 798). The last column is M3's hazard ratio per unit, $\exp(\hat{\beta} + \hat{\beta}^{\mathrm{tt}} g(t))$, at those two times. The p-values are not valid tests, since the covariates were selected from M0.

1 (Appendix B.11). Only the early interval excludes 1, so the later rise is uncertain. Like the p-values, these intervals ignore that the covariates were chosen from M0.

The global test does not reject proportional hazards, but one covariate's effect may reverse over follow-up, in which case a single hazard ratio misdescribes it.

4.5.2 Assumption 2: do the covariates enter as straight lines?

M1 and M2b replace each eligible continuous covariate with a restricted cubic spline of three degrees of freedom (Section 3.7.1). Eighteen of the twenty-seven pass the eligibility check (Appendix B.7) in every training fold, and the other nine stay linear in all four models (Table 4.17). Each spline adds two parameters, so the likelihood ratio test has $18 \times 2 = 36$ degrees of freedom (Appendix B.10). The test is run once under each link (Figure 3.8).

Comparison	Link	χ^2	df	p
M0 against M1	Cox	109.85	36	2.1×10^{-9}
M2a against M2b	Complementary log-log	111.61	36	1.1×10^{-9}

Table 4.11: Likelihood ratio tests of linear against spline models. The 5% critical value of χ^2_{36} is 51.0.

Linearity is rejected under both links, with nearly identical statistics of 109.85 and 111.61, each about twice the 5% critical value of 51.0 (Table 4.11).

The rejection could instead come from Efron's approximation, which is least reliable under ties as heavy as these (Section 4.3). Comparing M0 and M2a checks this. Both estimate the same β (2.3), but M2a handles ties exactly, so any difference between them comes from the approximation.

Comparison of the 27 coefficients	Value
Correlation between M0 and M2a coefficients	0.99999
Largest difference for any covariate	4.4×10^{-3}

Table 4.12: Agreement between M0 and M2a coefficients.

The coefficient vectors correlate at 0.99999 (Table 4.12, Figure 4.8). M2a and M2b, which use no tie approximation, also reject linearity about as strongly as M0 and M1 (Table 4.11). Efron's approximation is not driving the result.

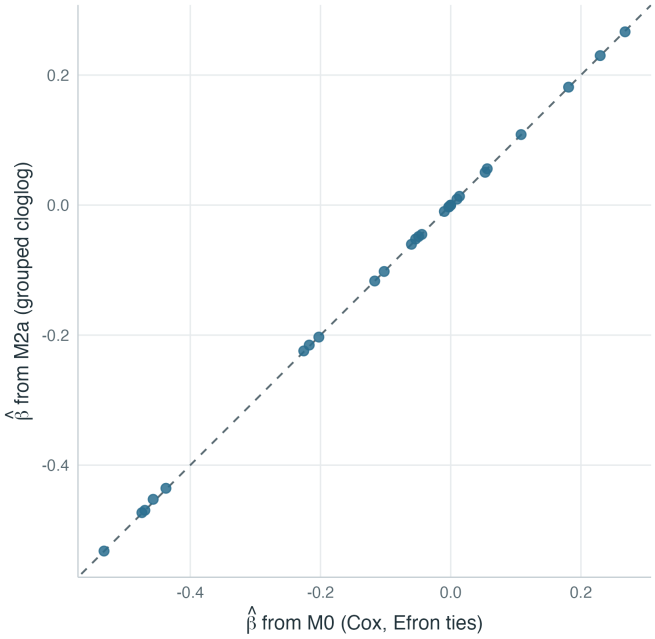


Figure 4.8: M0 against M2a coefficients. Points on the diagonal are unchanged by the tie approximation.

Tested one at a time, four of the eighteen splined covariates reject linearity at the 5% level, and only mean mileage survives a Bonferroni correction (Table B.4). Figure 4.9 shows these four against M0's linear term.

For mileage, vehicle age spread and days since the last new vehicle, the hazard at the low end is higher than the line allows, by about 0.3, 0.15 and 0.4 on the log scale. At the top of their range, mileage and vehicle age spread stop lowering the hazard while the line keeps falling. The usage concentration band

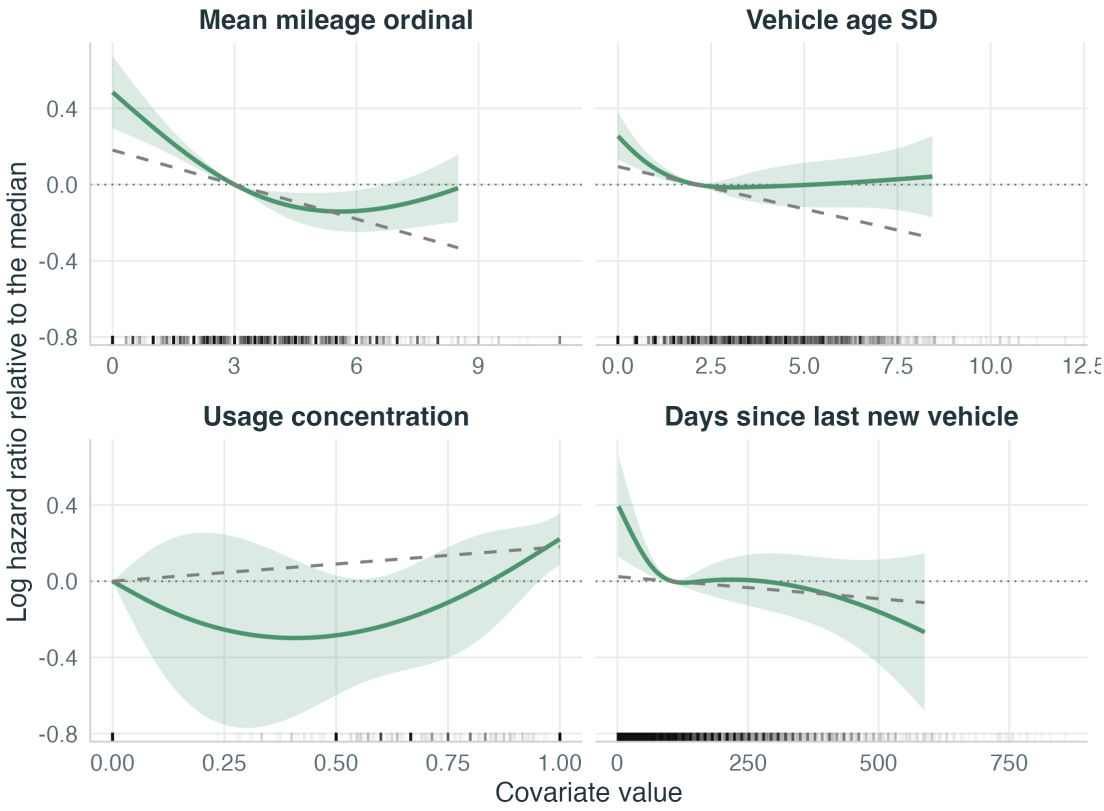


Figure 4.9: M1 spline (solid, with 95% cluster-robust band) against M0's linear term (dashed) for the four covariates that reject linearity on their own (Table B.4), both relative to the covariate's median. Ticks show a random sample of rows.

is too wide to tell the two apart. The departures sit at the ends of the ranges, which fits the small AUC gain (Section 4.5.3).

4.5.3 What relaxing the assumptions buys

To see whether the non-linearity also improves predictions for new customers (RQ2), each model except M3 is scored on held-out folds (Figure 3.10, Section 3.7.2). M4, which assumes neither proportional hazards nor linearity, shows how much a fully flexible model gains.

Model	AUC	Brier	Calibration slope
M0	0.6803 (0.0102)	0.021536 (0.000156)	0.919 (0.080)
M1	0.6822 (0.0099)	0.021534 (0.000152)	0.893 (0.080)
M2a	0.6804 (0.0108)	0.021536 (0.000156)	0.934 (0.081)
M2b	0.6826 (0.0106)	0.021533 (0.000152)	0.907 (0.079)
M4	0.6856 (0.0121)	0.021517 (0.000156)	0.981 (0.092)

Table 4.13: Out-of-fold performance, mean across five folds (standard deviation in brackets). Predicting the 2.22% base rate for every row gives a Brier score of 0.02174.

Comparison	Δ AUC	Δ Brier	Folds improved
M1 against M0	$+0.0019$	-0.000002	4 of 5
M2b against M2a	$+0.0022$	-0.000003	4 of 5
M2a against M0	$+0.0001$	≈ 0	3 of 5
M4 against M0	$+0.0053$	-0.000020	5 of 5
M4 against M1	$+0.0033$	-0.000017	4 of 5

Table 4.14: Paired fold-by-fold differences on the same folds. A negative Δ Brier is an improvement.

Relaxing linearity raises AUC by 0.0019 under the Cox link and 0.0022 under the complementary log-log link, improving four folds of five in both (Table 4.14). The Brier score does not separate the models, since all twenty-seven covariates together improve it by only 0.0002 over the base rate. So although linearity is rejected at $p < 10^{-8}$, relaxing it barely helps on unseen customers.

No two folds choose the same M4 configuration, although the settings stay close (Table 4.15). Relaxing both assumptions gives a larger gain. M4's AUC is on average 0.0053 higher than M0's, and higher in all five folds. It is on average 0.0033 higher than M1's, and higher in four folds of five (Table 4.14). Its Brier score is about 2×10^{-5} lower, ten times the improvement the splines gave. M4 also has the calibration slope closest to 1, at 0.981, against 0.89 to 0.93 for M0

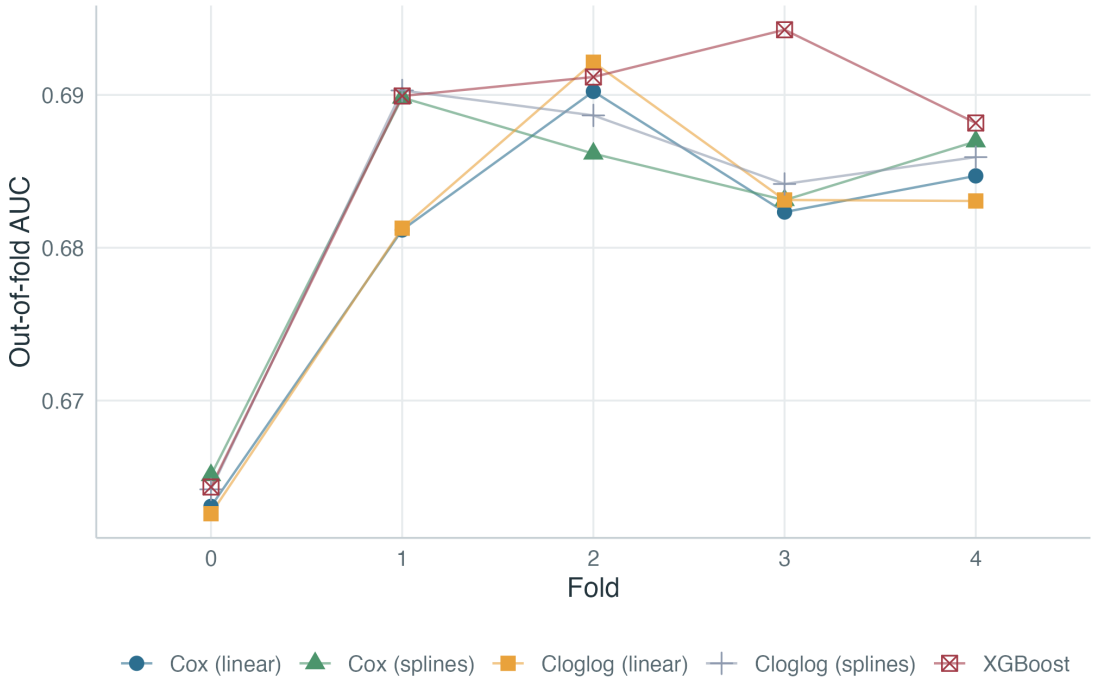


Figure 4.10: Out-of-fold AUC by fold.

Fold	Max depth	Learning rate	Rounds	Min child weight	Row subsample	Column subsample	L_2
0	3	0.05	200	5	1.0	1.0	5
1	4	0.03	400	1	0.6	0.6	10
2	3	0.05	200	5	0.6	0.6	10
3	4	0.05	200	3	0.8	0.6	5
4	3	0.05	100	1	0.6	1.0	1

Table 4.15: Selected XGBoost hyperparameters per outer fold, by mean inner-fold log loss.

to M2b (Table 4.13). The slopes vary between folds with a standard deviation of 0.079 to 0.092, as large as M4's lead, so its calibration advantage is less firm than in AUC, where it beats M0 in every fold (Figure 4.10). Fold 0 is about 0.02 lower for every model, so the paired differences in Table 4.14 are the fairer comparison.

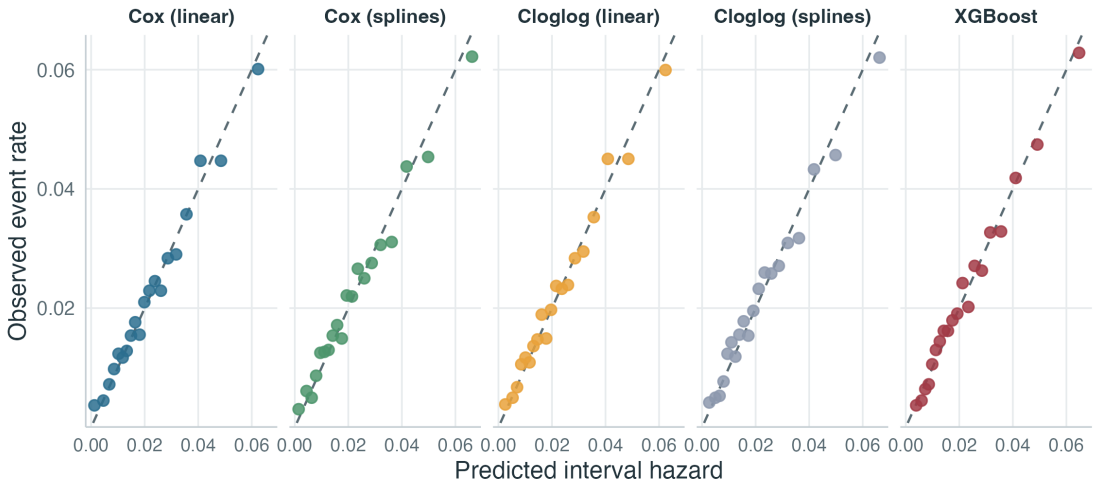


Figure 4.11: Predicted against observed hazard in twenty bins. At the top end, M0 to M2b sit below the diagonal, predicting more churn than occurs, and M4 does not.

4.5.4 What M4 relies on

SHAP values attribute each prediction to the covariates behind it [23]. They are computed out of fold, so each row is explained by a model that never trained on it.

The interval index has by far the largest mean |SHAP|, at 0.322 (Figure 4.12), more than twice the next covariate and about a fifth of the total across all twenty-eight inputs. M4 uses the interval index as a stand-in for the baseline hazard (Section 3.7.2). M0 has no such covariate, because it handles time through the baseline hazard $\lambda_0(t)$, so the interval index is left out of the comparison below. Table 4.16 lists the ten covariates M4 relies on most among the twenty-seven it shares with M0.

M4 and M0 largely agree on which covariates matter. All six of M0's most significant covariates are in M4's top ten by SHAP (Table 4.16). The other four, gap SD, gap mean, days since last new vehicle and actions-per-session drift, are not significant in M0 yet account for about a quarter of M4's total mean |SHAP|. All four are splined in M1 and M2b (Table 4.17), yet the splines added little AUC (Section 4.5.3). Figure 4.13 shows the direction of each effect.

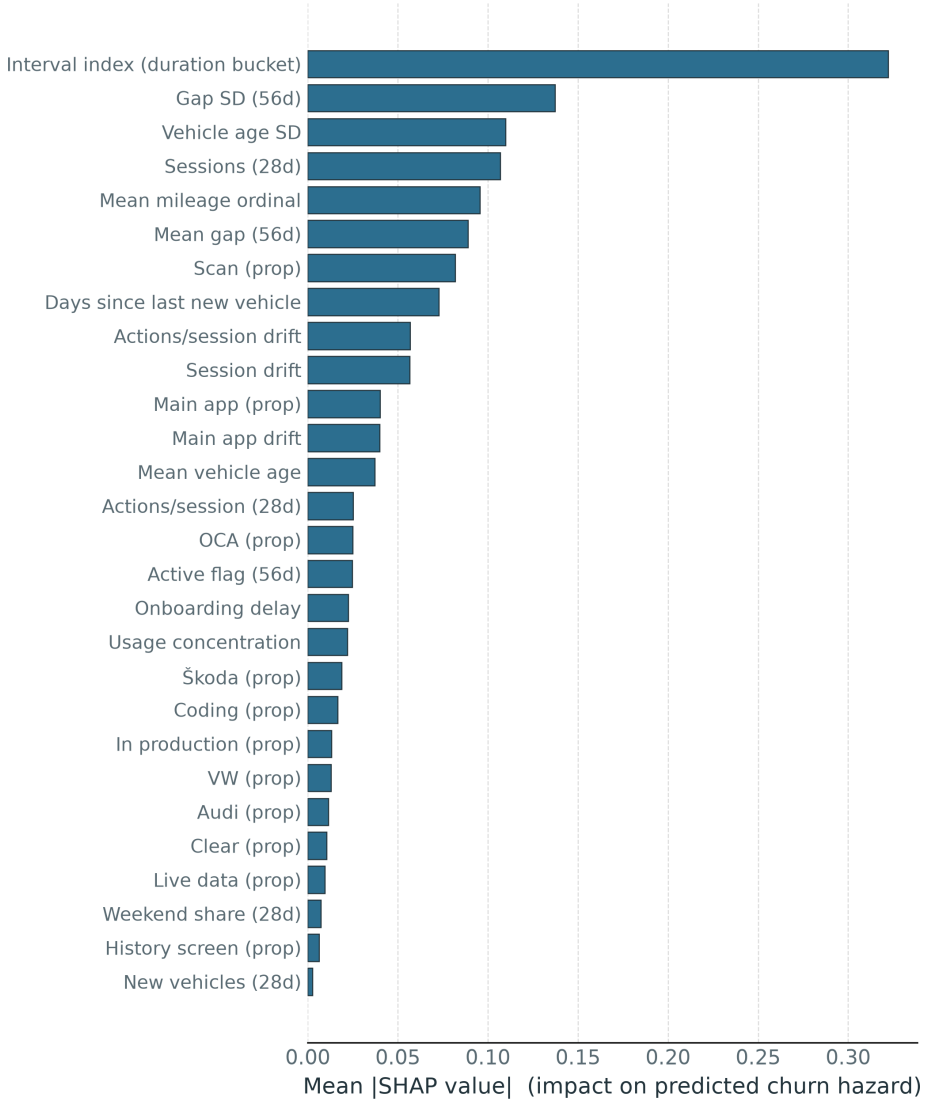


Figure 4.12: Mean absolute out-of-fold SHAP value per covariate.

Covariate	Mean |SHAP|	M0 rank by $\|z\|$	M0 p
Gap SD (56d)	0.137	15	0.065
Vehicle age SD	0.110	6	< 0.001
Sessions (28d)	0.107	1	< 0.001
Mean mileage ordinal	0.096	5	< 0.001
Gap mean (56d)	0.089	22	0.430
Scan proportion	0.082	2	< 0.001
Days since last new vehicle	0.073	21	0.395
Actions-per-session drift	0.057	27	0.868
Session drift	0.057	3	< 0.001
Main-app proportion	0.040	4	< 0.001

Table 4.16: The ten shared covariates M4 relies on most, against their standing in M0. M0 rank is by absolute test statistic, with 1 the strongest.

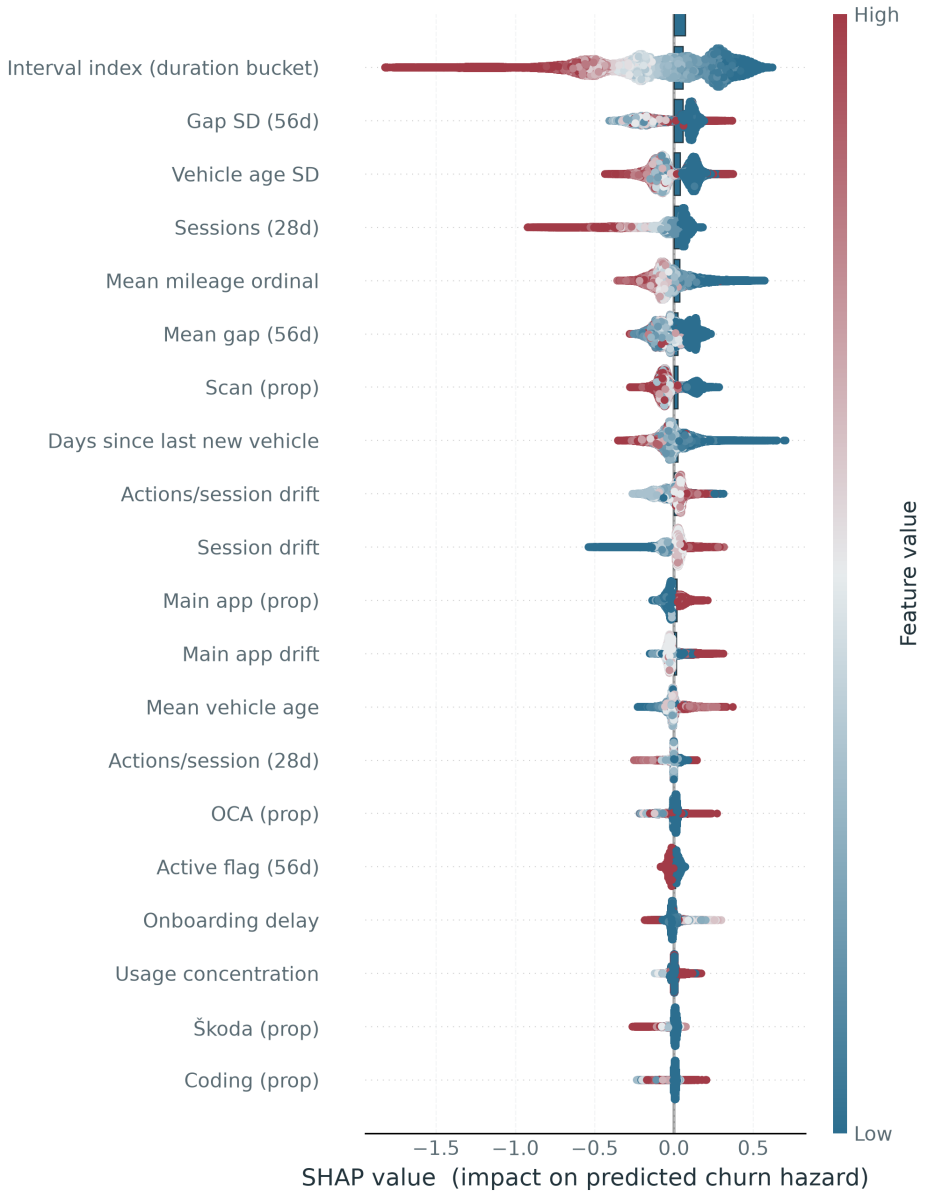


Figure 4.13: Out-of-fold SHAP values by covariate, coloured by the row's covariate value. High values to the right of zero mean large values raise the hazard.

4.5.5 What M0 estimates

Covariate	$\hat{\beta}$	HR per SD	95% CI	p	Splined
Sessions (28d)	-0.2025	0.556	0.477–0.648	< 0.001	Yes
Scan proportion	-0.4697	0.840	0.801–0.880	< 0.001	Yes
Session drift	$+0.1081$	1.319	1.208–1.439	< 0.001	Yes
Main-app proportion	$+0.2296$	1.108	1.059–1.158	< 0.001	Yes
Mean mileage ordinal	-0.0603	0.896	0.853–0.942	< 0.001	Yes
Vehicle age SD	-0.0444	0.900	0.857–0.945	< 0.001	Yes
One-click app proportion	-0.4374	0.924	0.886–0.964	< 0.001	No
History-screen proportion	-0.5324	0.931	0.894–0.970	0.001	No
Clear proportion	-0.4743	0.938	0.900–0.978	0.003	Yes
Usage concentration	$+0.1810$	1.088	1.026–1.153	0.005	Yes
Coding proportion	-0.4570	0.943	0.900–0.988	0.014	No
In-production proportion	$+0.2678$	1.041	1.008–1.076	0.014	No
Mean vehicle age	$+0.0135$	1.061	1.009–1.116	0.021	Yes
Škoda proportion	-0.2173	0.950	0.908–0.994	0.027	No
Gap SD (56d)	-0.0098	0.951	0.901–1.003	0.065	Yes
Live-data proportion	-0.2257	0.961	0.918–1.005	0.084	No
Active flag (56d)	-0.1167	0.945	0.874–1.020	0.147	No
Audi proportion	-0.1022	0.962	0.912–1.015	0.160	Yes
Main-app drift	$+0.0561$	1.027	0.984–1.072	0.226	Yes
New vehicles (28d)	$+0.0529$	1.029	0.976–1.084	0.289	No
Days since last new vehicle	-0.0002	0.971	0.906–1.040	0.395	Yes
Gap mean (56d)	-0.0028	0.977	0.921–1.036	0.430	Yes
Volkswagen proportion	-0.0539	0.980	0.929–1.033	0.446	Yes
Weekend share (28d)	-0.0493	0.985	0.943–1.029	0.497	No
Actions per session (28d)	$+0.0096$	1.016	0.930–1.110	0.723	Yes
Onboarding delay	-0.00005	0.997	0.965–1.030	0.843	Yes
Actions-per-session drift	-0.0030	0.994	0.923–1.070	0.868	Yes

Table 4.17: M0 with cluster-robust standard errors, ordered by p-value, with a rule at $p = 0.05$. Hazard ratios are per standard deviation. The last column marks the eighteen covariates splined in M1 and M2b.

Of its 27 covariates, 14 are significant at the 5% level (Table 4.17, Figure 4.14). One SD more sessions in the last 28 days lowers the hazard by 44%, the largest effect, and one SD more session drift raises it by 32%. The activity-mix proportions sum to 1, so each coefficient is the effect of moving activity from the omitted *other* group to that category. Moving activity to scan, clear, coding, history, or one-click lowers the hazard, and shifting activity from the VAG app to the main app raises it.

Session drift's hazard ratio of 1.32 seems to say that more activity raises churn, but it is measured at a fixed recent session count. Of two customers with 10 sessions in the last 28 days, the one who had 4 in the 28 days before has the

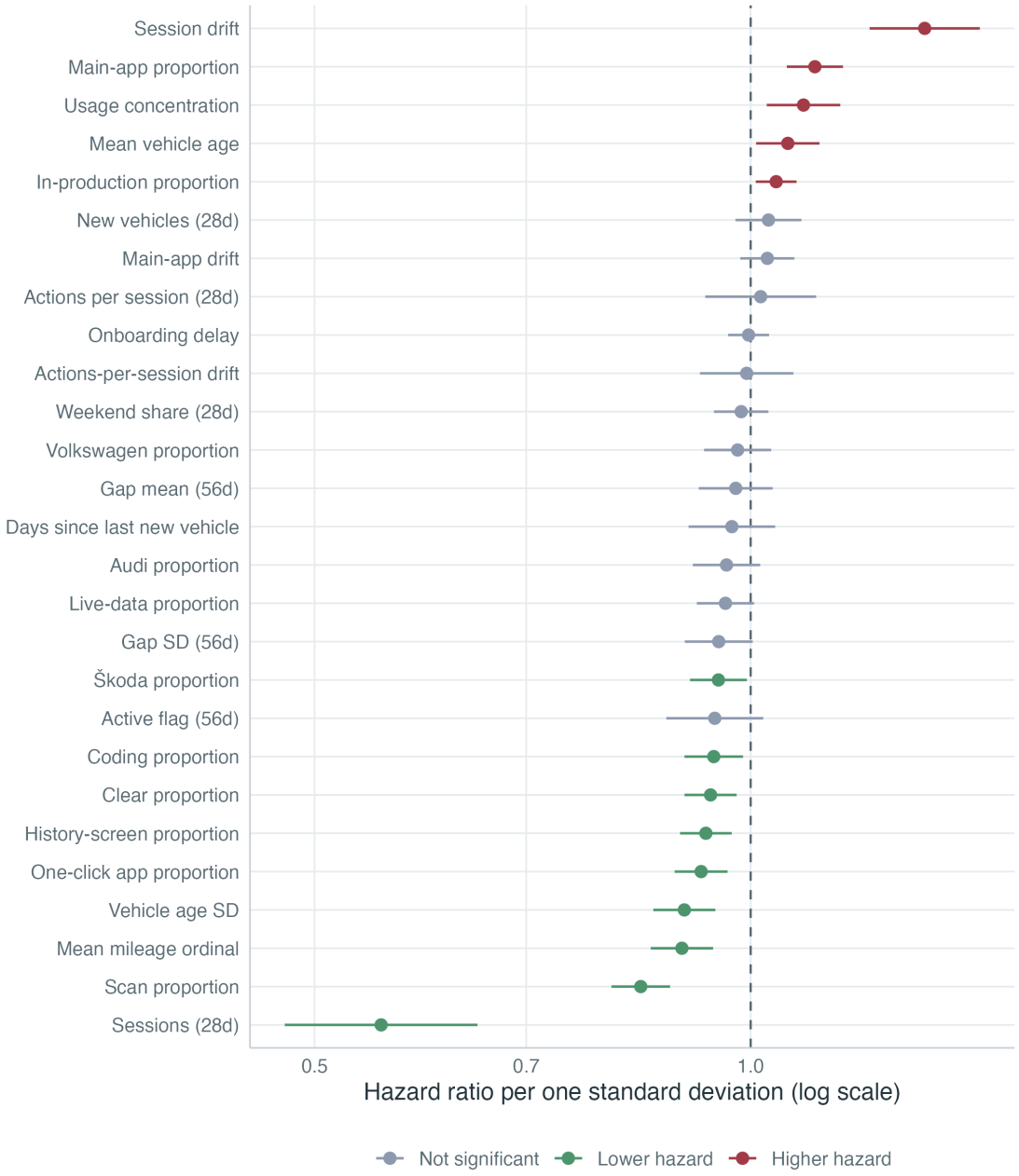


Figure 4.14: M0 hazard ratios per standard deviation, with 95% cluster-robust intervals.

higher hazard, so a recent rise to a given level signals more risk than a steady one.

Block	Covariates	Significant
Volume	Sessions (28d), session drift, actions per session (28d), actions-per-session drift	2 of 4
Activity mix	Scan, main-app, one-click app, history-screen, clear, coding and live-data proportions, usage concentration, main-app drift	7 of 9
Fleet	Mean mileage ordinal, mean vehicle age, vehicle age SD, Škoda, Audi, Volkswagen and in-production proportions, new vehicles (28d), days since last new vehicle	5 of 9
Timing	Gap SD (56d), gap mean (56d), weekend share (28d), onboarding delay, active flag (56d)	0 of 5

Table 4.18: Covariates by block, with the number significant at the 5% level in M0 (Table 4.17).

Volume has the two largest effects, and activity mix has the most significant covariates, 7 of 9 (Table 4.18). In the fleet block, mileage, vehicle age, the spread of vehicle ages, the Škoda share and the share of vehicles still in production are significant, while the two measures of adding vehicles are not. No timing covariate is significant in M0, although both gap covariates rank highly in M4 (Section 4.5.4).

5

Discussion

5.1 Answers to the research questions

This thesis tested the two assumptions of the Cox model, proportional hazards and linearity, and asked whether relaxing them improves predictions for new customers.

The two assumptions give different answers (RQ1). The global test does not reject proportional hazards under any transform, but the effect of the one-click app proportion may reverse during follow-up. Under the identity transform, its hazard ratio per SD moves from 0.890 at the first event time to 1.105 at the last, crossing 1 at about day 455. Linearity is rejected under both links ($\chi^2 = 109.85$ and 111.61 on 36 degrees of freedom, Table 4.11), and the tie check rules out Efron's approximation as the cause. Four covariates reject on their own at the 5% level, mostly at the low end of their ranges, although only mean mileage survives a Bonferroni correction (Figure 4.9, Table B.4). Asserting both without testing, as most of the reviewed churn studies do, would have missed both findings.

The results also go against Meža, Strle, and Meža [26], who state that both assumptions fail but report no test of either. In the OBDeleven data, proportional hazards is not rejected globally, although one covariate's effect may drift, while linearity is rejected under both links.

The non-linearity is real but matters little for prediction (RQ2). Splines raise out-of-fold AUC by 0.0019 under the Cox link and 0.0022 under the complementary log-log link. M4, which relaxes both assumptions, raises mean AUC by 0.0053 over M0 and beats it in all five folds. Its calibration slope is closest to 1, at 0.981 against 0.89 to 0.93 for M0 to M2b, although the variation between folds is as large as that lead (Table 4.13). All models discriminate only moderately, with AUCs between 0.680 and 0.686. Relaxing proportional hazards alone was not scored out of fold, so RQ2 is answered for linearity and for both assumptions together (Section 5.4).

Several results match the studies in Chapter 2. The one-click effect may be the case Stensrud and Hernán [38] describe, where M0's single hazard ratio of 0.924 per SD averages an early protective effect and a later rise. The global test missed it and the per-covariate test flagged it, as Park and Hendry [28] warn. The tie check is consistent with the simulation of Tutz and Schmid [45], since Efron's approximation and the grouped model give almost the same coefficients at 14-day intervals. None of the churn studies reviewed tests both assumptions on covariates updated over follow-up and scores its models on customers not used in fitting (Table 2.2).

In practice, both assumptions should therefore be tested whenever a Cox model's coefficients are used to explain churn.

5.2 Possible sources of M4's gain

The interval index carries more weight in M4's predictions than any other input, and four covariates that M0 cannot distinguish from noise (gap SD, gap mean, days since last new vehicle and actions-per-session drift) account for about a quarter of M4's total mean |SHAP|.

M4's small gain has no clear source. Possible causes are non-linearity the splines miss, effects that change over time (which a tree captures by splitting on both the interval index and a covariate), and interactions between covariates. Refitting M4 with trees of depth 1, which allow no interactions, would show how much of the gain is non-linearity alone.

5.3 Implications for OBDeleven

The models identify which customers are at risk, but they do not show what causes churn (Section 5.4). The points below are therefore signals to monitor and ideas to test.

Recent activity is the strongest signal in M0. A customer with one SD fewer sessions in the last 28 days has a hazard about 1.8 times higher. A drop in session count is the simplest early warning, and it can be tracked without a model.

At the same recent session count, customers whose activity has recently risen are at higher risk (session drift, HR 1.32 per SD). These may be customers returning after a quiet period who have not yet settled into regular use, and may be worth contacting.

Customers who use the diagnostic features (scan, clear, coding, history and one-click apps) have a lower churn hazard than customers whose activity goes to the pooled other types. Testing whether steering users towards them reduces churn would need an experiment, such as an A/B test. For one-click apps the lower hazard may hold only early in follow-up, since the hazard ratio crosses 1 at about day 455 (Section 4.5.1). One possible reason is that one-click apps make one-off changes to a car, so once a customer has made the changes they want, they have less reason to return. Customers who rely on the main app rather than the VAG app have a higher hazard. Adding vehicles is not linked to churn once the other covariates are held fixed, while the make-up of the fleet is (Table 4.18).

M4 ranks customers best, but its AUC is only 0.005 above M0's, which would probably change little in which customers a retention campaign targets. M0 is simpler and gives effect estimates with confidence intervals, though these must be read with the rejected linearity and the possible one-click reversal in mind. M4's SHAP values only show how the model uses each input. If the predicted probabilities themselves are used, for example to estimate how many customers will churn in the next 14 days, M0 to M2b overestimate risk for the customers they rank highest (Figure 4.11), so M4 or a recalibrated M0 should be used.

5.4 Scope and limitations

The limitations below concern internal validity, generalisability and the reliability of the tests, in that order.

Not causal. The coefficients show associations with churn, not causes. For example, if a price rise makes customers use the app less before they leave, the model credits the drop in activity and not the price rise.

Imputation leak. The imputer for `vehicle_start_year` is fitted on all the data, so a little information from the test folds reaches one covariate.

Constructed event. Churn is defined by an inactivity threshold, not an observed cancellation, and a different threshold would give different events, so the linearity rejection and the possible one-click reversal are conditional on $\theta = 160$ days.

Recurrent events. Only the first churn is modelled, so customers who leave and return are not studied.

Left truncation. All results apply only to customers who stayed for at least 56 days, so the models cannot flag new customers who leave in their first eight weeks.

One segment. Only personal users of one app are studied.

Segmentation. No true label exists, so some professional customers may remain in the personal segment. The rule also reads each customer's whole log, so a customer who stays longer can pass $q = 11$ vehicles and must then meet condition A or B, the grouping route to immortal-time bias (Section 2.3.2). Only customers above q are affected.

Fixed splines. Every spline puts its knots at the same percentiles, which may suit some covariates better than others, and nine covariates that are mostly zero or binary stay linear.

Few time interactions. M3 covers only three covariates and is not scored out of fold, so M4 is only compared with models whose effects are constant over time.

No sensitivity analysis. The churn threshold, the segmentation cut-offs and the interval width are each set once, and the models are not refitted at other values. The burst threshold was varied only for its effect on the segmentation.

5.5 Further work

Splines for mostly-zero covariates. An indicator for non-zero values plus a spline on the positive values would let the mostly-zero linear covariates be tested for linearity.

Tuned knots. Knot percentiles could be tuned inside each training fold, like M4's hyperparameters.

Additive models. Penalised splines would avoid choosing knots at all, and would sit between M1 and M4 in the ladder.

Joint models. Modelling each customer's covariate trajectory together with churn [41] would allow survival to be predicted forward.

Sensitivity analysis. Refitting M0 and M1 at one threshold below 160 days and one above, within the 86–213 days of Section 3.3, would show whether the linearity rejection and the possible one-click reversal depend on the event definition.

Scoring M3. Scoring M3 on the same folds would show whether relaxing proportional hazards alone improves prediction.

Leakage check. Refitting the models without the two vehicle-age covariates, or with the imputer fitted inside each training fold, would show whether the imputation leak changes the results.

5.6 Ethical and societal implications

The models give each customer a churn risk. If OBDeleven uses these scores to decide who gets retention offers, customers are treated differently based on app usage, so the terms of use should cover that. The coefficients are also not causal, so any app changes based on them should be tested first, for example in an A/B test. The wider societal impact is small, since the data are pseudonymous and only aggregate results are reported.

The AI tools listed in Section 3.7 could produce plausible but incorrect code or text, so their suggestions were checked against the sources, the code's output and the results before use. They did not produce data or results.

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A

Notation & Derivations

This appendix collects the notation, formulae, and derivations behind the results stated in Chapter 2, in the order in which that chapter uses them. The subscript i is suppressed where a derivation concerns a single subject.

A.1 Notation

Table A.1 lists the notation used in Chapter 2 and in this appendix.

Symbol	Meaning
i	subject index, $i = 1, \ldots, n$
j	interval index, $j = 1, \ldots, J$
k	covariate index, $k = 1, \ldots, p$
t_j	end of interval j, so that interval j is $(t_{j-1}, t_j]$ and $t_0 = 0$
J_i	last interval in which subject i is observed
T	event time
C	censoring time
$\tilde{T}$	observed time, $\tilde{T} = \min(T, C)$
τ	horizon: fixed cutoff defining a binary outcome
Y	binary outcome, $Y = \mathbf{1}\{T \leq \tau\}$
x_{ij}	covariate vector for subject i in interval j
d_{ij}	event indicator: 1 if subject i churns in interval j, else 0
d_i	event indicator for subject i: 1 if churned, 0 if censored
d_j	number of events at t_j, $d_j = \sum_i d_{ij}$
n_j	number at risk immediately before t_j
h_{ij}	discrete hazard for subject i in interval j
γ_j	baseline intercept for interval j
Λ_{0j}	baseline cumulative hazard over interval j, $\Lambda_{0j} = e^{\gamma_j}$
$\lambda_0(t)$	baseline hazard
β	vector of regression coefficients
z_i	frailty, an unobserved subject-specific multiplier on the hazard
$\lambda(t)$	continuous hazard
$\Lambda(t)$	cumulative hazard, $\Lambda(t) = \int_0^t \lambda(u)\, du$
$S(t)$	survival function, $S(t) = P(T > t)$
$f(t)$	density (mass) of the event at time t
$F(t)$	distribution function, $F(t) = P(T \leq t)$
$f_C(t)$	density of the censoring time
$F_C(t)$	distribution function of the censoring time, $F_C(t) = P(C \leq t)$
$S_C(t)$	survival function of the censoring time, $S_C(t) = P(C > t)$
$R(t)$	risk set at t: subjects under observation and event-free
$N_i(t)$	number of events observed for subject i over $[0, t]$
$Y_i(t)$	at-risk indicator: 1 if subject i is under observation and event-free just before t (not the outcome Y)

Table A.1: Notation used throughout.

A.2 Basic quantities

The basic quantities follow Kalbfleisch and Prentice [18, ch. 1]:

$f(t)$	density (mass) of the event at time t
$F(t) = P(T \le t)$	probability of the event occurring by time t
$S(t) = P(T > t)$	probability of no event by time t
$\lambda(t)$	continuous hazard – a *rate*, bounded below by zero, unbounded above
$h(t)$	discrete hazard – a *probability*, confined to $[0, 1]$ *(this work)*

These functions are related as follows. The distribution function accumulates the density, and the survival function is its complement:

$$F(t) = \int_0^t f(u)\,du, \qquad S(t) = 1 - F(t). \tag{A.1}$$

The continuous hazard is defined as the limit, as $\Delta t \to 0$, of the conditional probability of the event in $[t, t + \Delta t)$ given survival to t, divided by Δt:

$$\lambda(t) = \lim_{\Delta t \to 0} \frac{P(t \le T < t + \Delta t \mid T \ge t)}{\Delta t}. \tag{A.2}$$

Its relation to f and S follows from the definition of conditional probability. Writing $A = \{t \le T < t + \Delta t\}$ and $B = \{T \ge t\}$, and noting that $A \subseteq B$, so that $A \cap B = A$:

$$P(t \le T < t + \Delta t \mid T \ge t) = \frac{P(t \le T < t + \Delta t)}{P(T \ge t)} = \frac{F(t + \Delta t) - F(t)}{S(t)}. \tag{A.3}$$

Substituting into the definition of $\lambda(t)$, and noting that $S(t)$ does not depend on Δt and may therefore be taken outside the limit:

$$\lambda(t) = \lim_{\Delta t \to 0} \frac{1}{\Delta t} \cdot \frac{F(t + \Delta t) - F(t)}{S(t)} = \frac{1}{S(t)} \lim_{\Delta t \to 0} \frac{F(t + \Delta t) - F(t)}{\Delta t} = \frac{f(t)}{S(t)}, \tag{A.4}$$

where the remaining limit is the derivative of F, which is the density f. For continuous T the event $\{T = t\}$ has probability zero, so $P(T \ge t) = P(T > t) = S(t)$ and the distinction is immaterial; this ceases to hold once time is discretised.

As previously specified, $\lambda(t)$ is strictly a *rate* and can be read as a probability only approximately, and only for small Δt, via equation (A.5)

$$P(t \leq T < t + \Delta t \mid T \geq t) \approx \lambda(t)\,\Delta t \tag{A.5}$$

The hazard may equivalently be written as a derivative of the log survival function. Applying the chain rule to $\log S(t)$ and using $S'(t) = -f(t)$ from (A.1):

$$\frac{d}{dt}\log S(t) = \frac{S'(t)}{S(t)} = -\frac{f(t)}{S(t)} = -\lambda(t). \tag{A.6}$$

Integrating both sides from 0 to t and applying the fundamental theorem of calculus:

$$\int_0^t \lambda(u)\,du = -\int_0^t \frac{d}{du}\log S(u)\,du = -\big[\log S(t) - \log S(0)\big]. \tag{A.7}$$

Since no subject has experienced the event at the time origin, $S(0) = 1$ and $\log S(0) = 0$. Defining the cumulative hazard $\Lambda(t) = \int_0^t \lambda(u)\,du$ then gives

$$\Lambda(t) = -\log S(t), \qquad S(t) = e^{-\Lambda(t)}, \qquad f(t) = \lambda(t)e^{-\Lambda(t)}. \tag{A.8}$$

A.3 The likelihood

The likelihood combines both the density of failing and the probability of no event at time t

$$L = \prod_i f(t_i)^{d_i}\, S(t_i)^{1-d_i}. \tag{A.9}$$

Substituting Equations (A.4) and (A.8) into the above gives the likelihood expressed in terms of the hazard function

$$L = \prod_i \big[\lambda(t_i)S(t_i)\big]^{d_i} S(t_i)^{1-d_i} = \prod_i \lambda(t_i)^{d_i}\, S(t_i) = \prod_i \lambda(t_i)^{d_i}\, e^{-\Lambda(t_i)}. \tag{A.10}$$

Equation (A.9) leaves out the censoring times. A subject with an event at t_i was not censored before it, and a subject censored at t_i had not failed, so with f_C and S_C the density and survival function of C, the full likelihood is

$$L = \prod_i \big[f(t_i)\, S_C(t_i)\big]^{d_i} \big[S(t_i)\, f_C(t_i)\big]^{1-d_i} = \prod_i f(t_i)^{d_i}\, S(t_i)^{1-d_i} \times \prod_i S_C(t_i)^{d_i}\, f_C(t_i)^{1-d_i}. \tag{A.11}$$

If C is independent of T given the covariates and f_C and S_C share no parameters with f, the second product does not depend on the parameters of interest and can be dropped when maximising, which returns Equation (A.9) [18, ch. 6]. This is the uninformative censoring condition.

A.4 The Cox model

Under (2.1), the ratio of hazards for any two subjects

$$\frac{\lambda(t \mid x_i)}{\lambda(t \mid x_j)} = \frac{\lambda_0(t)\exp(x_i'\beta)}{\lambda_0(t)\exp(x_j'\beta)} = \exp\big((x_i - x_j)'\beta\big) \tag{A.12}$$

does not depend on t, since the baseline is shared by all subjects and cancels.

The partial likelihood contribution is the probability that the event at t_i belonged to subject i rather than to any other member of the risk set $R(t_i)$,

$$\frac{\lambda_0(t_i)\exp(x_i'\beta)}{\sum_{j \in R(t_i)} \lambda_0(t_i)\exp(x_j'\beta)} = \frac{\exp(x_i'\beta)}{\sum_{j \in R(t_i)} \exp(x_j'\beta)}, \tag{A.13}$$

where $R(t_i)$ is the set of subjects under observation just before t_i. The baseline appears in every term of the numerator and denominator and therefore cancels. Multiplying over all event times gives the partial likelihood [6]

$$L(\beta) = \prod_{i:\, d_i = 1} \frac{\exp(x_i'\beta)}{\sum_{j \in R(t_i)} \exp(x_j'\beta)}. \tag{A.14}$$

A.5 Grouped and discrete time

A.5.1 The discrete hazard

Define the discrete hazard for interval j as the conditional probability of the event occurring within $(t_{j-1}, t_j]$ given survival to its start [45, §3.3.2]:

$$h_j = P\big(t_{j-1} < T \le t_j \mid T > t_{j-1}\big) = 1 - \frac{S(t_j)}{S(t_{j-1})}. \tag{A.15}$$

The survival function at the end of the interval is the probability mass accumulated in that window subtracted from the survival at its start,

$$S(t_j) = S(t_{j-1}) - \int_{t_{j-1}}^{t_j} f(u)\,du, \tag{A.16}$$

and dividing through by $S(t_{j-1})$ recovers (A.15), since the integral equals $S(t_{j-1}) - S(t_j)$:

$$\frac{S(t_{j-1}) - S(t_j)}{S(t_{j-1})} = \frac{1}{S(t_{j-1})} \int_{t_{j-1}}^{t_j} f(u)\, du = h_j. \tag{A.17}$$

A.5.2 The complementary log-log model

Turning to the ratio $S(t_j)/S(t_{j-1})$, which is $1 - h_j$ by (A.15), and substituting $S(t) = e^{-\Lambda(t)}$ from (A.8), together with the proportional hazards form (2.1):

$$\frac{S(t_j)}{S(t_{j-1})} = \frac{e^{-\Lambda(t_j)}}{e^{-\Lambda(t_{j-1})}} = \exp\big(-[\Lambda(t_j) - \Lambda(t_{j-1})]\big) = \exp\left(-\int_{t_{j-1}}^{t_j} \lambda_0(u) e^{x'\beta}\, du\right). \tag{A.18}$$

Within a given interval the covariate vector is held constant [43], so $e^{x'\beta}$ does not depend on u and may be taken outside the integral:

$$1 - h_j = \exp\left(-e^{x'\beta} \int_{t_{j-1}}^{t_j} \lambda_0(u)\, du\right). \tag{A.19}$$

Writing $\gamma_j = \log \int_{t_{j-1}}^{t_j} \lambda_0(u)\, du$ for the accumulated baseline over the interval gives

$$1 - h_j = \exp\big(-e^{\gamma_j + x'\beta}\big), \tag{A.20}$$

and taking logarithms twice yields

$$\log\big(-\log(1 - h_j)\big) = \gamma_j + x'\beta,$$

which is (2.3). Writing $\Lambda_{0j} = \int_{t_{j-1}}^{t_j} \lambda_0(u)\, du = e^{\gamma_j}$ for the baseline cumulative hazard over interval j,

$$-\log\big(1 - h_{ij}\big) = \Lambda_{0j}\, e^{x'_{ij}\beta}. \tag{A.21}$$

Rearranging (A.20) gives the hazard as a function of the linear predictor:

$$h_j = 1 - \exp\big(-\exp(\gamma_j + x'\beta)\big). \tag{A.22}$$

A.5.3 Survival as a product of conditional probabilities

Since $h_j = 1 - S(t_j)/S(t_{j-1})$ from (A.15), each interval contributes a conditional survival probability $1 - h_j = S(t_j)/S(t_{j-1})$. Writing $S(t_k)$ as a telescoping

product of such ratios, with $t_0 = 0$ and $S(t_0) = 1$:

$$S(t_k) = \prod_{j=1}^{k} \frac{S(t_j)}{S(t_{j-1})} = \frac{S(t_k)}{S(t_0)} = \prod_{j=1}^{k} \big(1 - h_j\big). \tag{A.23}$$

No independence assumption is used here: each h_j is by definition conditional on survival to the start of interval j, so the factors chain by the multiplication rule for conditional probabilities.

A.5.4 The person-period likelihood

The likelihood formulation follows from the following facts. For a subject censored in interval j,

$$L_i = (1 - h_1)(1 - h_2) \cdots (1 - h_j) = S(t_j), \tag{A.24}$$

whereas for a subject churning in interval j,

$$L_i = (1 - h_1)(1 - h_2) \cdots (1 - h_{j-1})\, h_j = S(t_{j-1})\, h_j. \tag{A.25}$$

Both cases are captured by a single expression, with $d_{ij} = 1$ if subject i experiences the event in interval j and 0 otherwise, since $h^0 = 1$ and $(1 - h)^0 = 1$ delete the inapplicable factor in each row:

$$L(\gamma, \beta) = \prod_{i=1}^{n} \prod_{j=1}^{J_i} h_j(x_{ij})^{\,d_{ij}} \, \big(1 - h_j(x_{ij})\big)^{1 - d_{ij}}, \tag{A.26}$$

where J_i is the last interval in which subject i is observed. Taking logarithms turns the product into a sum over all person-interval rows:

$$\ell(\gamma, \beta) = \sum_{i=1}^{n} \sum_{j=1}^{J_i} \Big[\, d_{ij} \log h_j(x_{ij}) + (1 - d_{ij}) \log \big(1 - h_j(x_{ij})\big) \Big], \tag{A.27}$$

which is maximised jointly over β and the interval intercepts $\gamma = (\gamma_1, \ldots, \gamma_J)$. The likelihood is exactly equivalent to that of a binomial model with independent observations d_{ij} with subject-specific probabilities h_j and covariate values x_{ij} [40]. The same person-period data may be fitted with a logistic link,

$$\log\!\left(\frac{h_j}{1 - h_j}\right) = \gamma_j + x'\beta. \tag{A.28}$$

A.6 The Kaplan–Meier estimator

Let $t_1 < t_2 < \cdots < t_J$ be the distinct times at which events are observed, n_j the number at risk immediately before t_j, and $d_j = \sum_i d_{ij}$ the number of events at t_j. Since h_j is the conditional probability of the event at t_j given survival to that point, the obvious estimate is the observed proportion among those at risk:

$$\hat{h}_j = \frac{d_j}{n_j}. \tag{A.29}$$

Substituting into (A.23) gives the Kaplan–Meier estimator [19]

$$\hat{S}(t) = \prod_{j:\, t_j \le t} \left(1 - \frac{d_j}{n_j}\right). \tag{A.30}$$

A.7 Classification under censoring

With a horizon τ fixed, the outcome label is defined by

$$Y = \mathbf{1}\{T \le \tau\}, \tag{A.31}$$

using only covariate information available at time zero. What is observed is $\tilde{T} = \min(T, C)$, where C is the censoring time, and $d = \mathbf{1}\{T \le C\}$, so Y is recoverable only in some cases:

$$Y = \begin{cases} 1, & d = 1,\ \tilde{T} \le \tau, \\ 0, & \tilde{T} > \tau, \\ \text{undefined}, & d = 0,\ \tilde{T} \le \tau. \end{cases} \tag{A.32}$$

The third case is undefined because $d = 0$ with $\tilde{T} \le \tau$ leaves two possibilities, $C < T \le \tau$ and $T > \tau$. The determined cases are an event observed by τ, with probability $P(T \le \tau,\, T \le C)$, and a subject observed beyond τ, with probability $P(T > \tau,\, C > \tau) = S(\tau)\, S_C(\tau)$ under independent censoring. Excluding the undetermined cases therefore replaces the estimand $F(\tau)$ with

$$\frac{P(T \le \tau,\, T \le C)}{P(T \le \tau,\, T \le C) + S(\tau)\, S_C(\tau)}, \tag{A.33}$$

which equals $F(\tau)$ when $F_C(\tau) = 0$, since every subject is then observed up to τ, the numerator is $F(\tau)$ and the denominator is $F(\tau) + S(\tau) = 1$. In general, it is at least $F(\tau)$. Under independent censoring the numerator is $\int_0^\tau f(u)\, P(C \ge$

$u)\,du$, which is at least $F(\tau)\,S_C(\tau)$ because $P(C \geq u) \geq S_C(\tau)$ for $u \leq \tau$, and a numerator $N \geq F(\tau)S_C(\tau)$ gives $N/\big(N+S(\tau)S_C(\tau)\big) \geq F(\tau)$, with equality only when $F_C(\tau) = 0$. Excluding the undetermined cases therefore over-represents those with the outcome.

A.8 Relative and additive risk models

Relative risk regression [2] is expressed as

$$\lambda(t \mid x_i) = \lambda_0(t)\, r\big(\beta, x_i(t)\big), \tag{A.34}$$

where $r(\beta, x(t))$ is the relative risk function. Cox's model is the case $r = \exp(x(t)'\beta)$, and other choices are the linear relative risk function $r = 1 + x(t)'\beta$ and the excess relative risk model $r = \prod_{k=1}^{p}\{1 + \beta_k x_k(t)\}$. In the Cox case, comparing two covariate vectors that differ by one unit in x_k alone multiplies the hazard by e^{β_k}, and every other covariate cancels. For the same comparison, the linear form gives $(1 + x'\beta + \beta_k)/(1 + x'\beta)$, which depends on every covariate, and the excess form gives $\big(1 + \beta_k(x_k + 1)\big)/(1 + \beta_k x_k)$, which depends on the value of x_k; only the Cox form gives a ratio free of the covariate values. With the study population grouped into S strata, each with its own baseline,

$$\lambda_s(t \mid x) = \lambda_{0s}(t) \exp\big(x(t)'\beta\big), \qquad s = 1, \ldots, S, \tag{A.35}$$

where β is common to all strata and $\lambda_{0s}(t)$ is left unspecified within each. The partial likelihood becomes a product of within-stratum partial likelihoods, since risk sets are formed only among subjects in the same stratum.

The additive model of Aalen [1] replaces the product with a sum,

$$\lambda(t \mid x_i) = \beta_0(t) + \beta_1(t)x_{i1}(t) + \cdots + \beta_p(t)x_{ip}(t). \tag{A.36}$$

A.9 The counting process

With $N_i(t)$ and $Y_i(t)$ as in Table A.1, the model specifies the intensity of the counting process,

$$E\big[dN_i(t) \mid \mathcal{F}_{t-}\big] = Y_i(t)\, \lambda_0(t) \exp\big(x_i(t)'\beta\big)\, dt, \tag{A.37}$$

where $\mathcal{F}_{t-}$ denotes everything observed before t.

With $\lambda_i(u) = \lambda_0(u)\exp\left(x_i(u)'\beta\right)$, the difference between the intensity and what happened is

$$M_i(t) = N_i(t) - \int_0^t Y_i(u)\,\lambda_i(u)\,du. \tag{A.38}$$

The integral is the compensator, the number of events predicted by time t. By construction, the next increment of M_i has expectation zero given everything observed before it, so M_i is a martingale [2, §2.2.5].

A.10 Immortal-Time Bias: A Constructed Example

The example below illustrates immortal-time bias (Section 2.3.2).

As an illustration, consider ten subjects with a constant discrete hazard of 0.2 per week that is *entirely unaffected* by enabling automations. Suppose every subject surviving to week four enables them (Table A.2).

subject	Churn week	Ever enabled automations
C1	1	No
C2	1	No
C3	2	No
C4	2	No
C5	3	No
C6	4	Yes
C7	5	Yes
C8	6	Yes
C9	8	Yes
C10	10 (censored)	Yes

Table A.2: Constructed example. The hazard is constant and independent of automations; the apparent effect is an artefact of when the feature can first be enabled.

Grouping by whether a subject ever enabled automations gives a mean survival of 1.8 weeks for non-users against 6.6 weeks for users. Automations appear to more than triple retention, when by construction they do nothing at all.

A.11 Frailty

Averaging the conditional hazard $z\,\lambda_0(t)e^{x_i'\beta}$ of a frailty model over the frailties of the subjects still event-free at t gives

$$\lambda(t \mid x_i) = \lambda_0(t)\, e^{x_i'\beta}\; E\big[z \mid T > t,\, x_i\big]. \tag{A.39}$$

The factor $E[z \mid T > t, x_i]$ changes with t and differs between covariate values, so the marginal hazard ratio drifts over time even though β does not.

B

Method Derivations

This appendix collects the formulae and derivations behind Chapter 3, in the order they are used there, followed by two used in Chapter 4. Appendix C contains the covariate table.

B.1 Effective number of vehicles

Let customer i have activity on V_i distinct vehicles after bursts are collapsed. Let a_{iv} be the number of actions on vehicle v, and let

$$p_{iv} = \frac{a_{iv}}{\sum_{w=1}^{V_i} a_{iw}}, \qquad \sum_{v=1}^{V_i} p_{iv} = 1 \tag{B.1}$$

be the share of the customer's activity on that vehicle. Concentration is measured by the Herfindahl–Hirschman index, the sum of squared shares [42],

$$\mathrm{HHI}_i = \sum_{v=1}^{V_i} p_{iv}^2. \tag{B.2}$$

Since $p_{iv}^2 \le p_{iv}$, the index is at most $\sum_v p_{iv} = 1$, reached when one share is 1. By Cauchy–Schwarz, $1 = \left(\sum_v p_{iv}\right)^2 \le V_i \sum_v p_{iv}^2$, so it is at least $1/V_i$, reached when every share is $1/V_i$:

$$\frac{1}{V_i} \le \mathrm{HHI}_i \le 1. \tag{B.3}$$

Taking the reciprocal flips the bounds and gives the effective number of vehicles,

$$N_i = \frac{1}{\mathrm{HHI}_i}, \qquad 1 \le N_i \le V_i. \tag{B.4}$$

A customer using N vehicles equally has shares $1/N$ and index $N \cdot (1/N)^2 = 1/N$. N_i is therefore the number of equally used vehicles that would produce the same concentration as the customer's actual uneven usage, and it

is smaller than V_i unless every share is equal. This reading is due to Laakso and Taagepera (1978, 1979), cited in Taagepera and Grofman [42]. Table B.1 gives worked values.

Usage pattern	HHI_i	N_i
All activity on one vehicle	1.00	1.00
Two vehicles, $50/50$	0.50	2.00
Three vehicles, $33/33/33$	0.33	3.00
Two vehicles, $80/20$	0.68	1.47
Three vehicles, $60/20/20$	0.44	2.27

Table B.1: Effective number of vehicles for a few usage patterns, from (B.2) and (B.4).

B.2 Segmentation rule

Let T_i be the four vehicles carrying the largest shares (all of them when $V_i < 4$); the top-four share is $C_{4i} = \sum_{v \in T_i} p_{iv}$. It equals 1 whenever $V_i \leq 4$, so customers with four or fewer vehicles automatically pass condition B. A customer is personal if

$$\text{personal}_i = \underbrace{\mathbf{1}\{N_i \leq \kappa\}}_{\text{A}} \vee \underbrace{\mathbf{1}\{C_{4i} \geq 0.80\}}_{\text{B}} \vee \underbrace{\mathbf{1}\{V_i \leq q\}}_{\text{C}}, \tag{B.5}$$

where $\kappa = 6$ is the threshold on the effective vehicle count and $q = 11$ is the 90th percentile of distinct vehicle counts (Section 4.1).

B.3 End-point bias from late entrants

Let r_t be the number at risk at t and s_t the number surviving beyond t, so the survival step at t is s_t/r_t. A customer entering within θ days of the cutoff cannot yet be labelled churned, so if kept, they enter both counts as survivors at each t where they are at risk. With m_t of them at risk at t the step becomes

$$\frac{s_t + m_t}{r_t + m_t}, \tag{B.6}$$

and

$$\frac{s_t + m_t}{r_t + m_t} - \frac{s_t}{r_t} = \frac{m_t\,(r_t - s_t)}{r_t\,(r_t + m_t)} \;\geq\; 0, \qquad \lim_{m_t \to \infty} \frac{s_t + m_t}{r_t + m_t} = 1. \tag{B.7}$$

Every step with an event is raised, and the product of the steps, the estimated survival function, is driven toward one as the late entrants grow in number. Meža, Strle, and Meža [26] describe this as end-point bias.

B.4 Splitting follow-up into intervals

Cut customer i's follow-up at the grid points $t_0 < t_1 < \cdots < t_{J_i}$, so row j covers $(t_{j-1}, t_j]$ and holds the covariate value x_{ij}. The intervals are disjoint, so exactly one row covers any time t, and the at-risk indicator of the customer is

$$Y_i(t) = \sum_{j=1}^{J_i} \mathbf{1}\{t_{j-1} < t \leq t_j\} \in \{0, 1\}.$$

The partial likelihood (A.14) is a product over event times t of

$$\frac{\exp\left(x_i(t)'\beta\right)}{\sum_l Y_l(t) \exp\left(x_l(t)'\beta\right)},$$

with i the customer failing at t. Each customer enters each denominator once, through the one row covering t, with the covariate value current at t. The split therefore returns the likelihood of the customer's covariate path, and since the log-likelihood is additive over time, nothing is counted twice [4].

B.5 Covariate windows and drift

With the recent and prior 28-day windows written $(0, 28]$ and $(28, 56]$, counted back from the end of interval j, and $x_{ij}^{(a,b]}$ the value of a covariate computed over window $(a, b]$, a drift covariate is

$$\text{drift}_{ij} = x_{ij}^{(0,28]} - x_{ij}^{(28,56]}. \tag{B.8}$$

With interval width w, a 28-day window spans $28/w$ rows and a 56-day window $56/w$. At $w = 14$, that is two and four rows, so (B.8) compares one two-row block against another. At $w = 28$, each 28-day window is a single row, and drift reduces to the difference between consecutive rows with no averaging behind it. A 56-day width was not used, since a 28-day window would then be half a row, and the frame holds nothing below interval resolution.

B.6 Entry point on the 14-day grid

Table B.2 shows the first rows of one customer on the 14-day grid.

Row starts	Windows close	Prior 28-day window (days)	Row asks about (days)	Status
0	14	-42 to -14	14 to 28	dropped
14	28	-28 to 0	28 to 42	dropped
28	42	-14 to 14	42 to 56	dropped
42	56	0 to 28	56 to 70	first modelled
56	70	14 to 42	70 to 84	modelled
70	84	28 to 56	84 to 98	modelled

Table B.2: The first rows of one customer on the 14-day grid. The first three fail on two counts. Their prior 28-day window reaches back before the customer's first recorded activity, and the event they ask about would fall before day 56, which the span filter has already ruled out. Day 42 is the first row where both conditions clear, so the exported frame begins there rather than at day 56.

B.7 Spline eligibility

Table B.3 illustrates the percentile condition of the eligibility check on two covariates.

	Ten sorted values	Minimum	67th percentile	Splined
Mostly zero	0, 0, 0, 0, 0, 0, 0, 0, 0.3, 0.5	0	0	no
Spread out	0, 0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8	0	0.5	yes

Table B.3: The spline-eligibility check on two covariates with the same minimum. The upper interior knot sits at the 67th percentile. In the first row, that percentile coincides with the minimum, so both interior knots land on the same value and the basis loses rank. In the second, it does not, and the covariate is splined.

B.8 Interval event probability for the Cox models

The primary model is

$$\lambda\big(t \mid x_i(t)\big) = \lambda_0(t) \exp\big(x_i(t)'\beta\big). \tag{B.9}$$

With x held constant on $(t_1, t_2]$, the probability of the event in that interval, given survival to t_1, follows as in Appendix A.5.2:

$$1 - \frac{S(t_2 \mid x)}{S(t_1 \mid x)} = 1 - \exp\Big(- e^{x'\beta} \int_{t_1}^{t_2} \lambda_0(u)\, du \Big).$$

Replacing the integral with the increment of the Breslow estimate $\hat{\Lambda}_0$ of the cumulative baseline hazard gives

$$\hat{h} = 1 - \exp\big(- \Delta\hat{\Lambda}_0 \exp(x'\hat{\beta})\big), \qquad \Delta\hat{\Lambda}_0 = \hat{\Lambda}_0(t_2) - \hat{\Lambda}_0(t_1). \tag{B.10}$$

$\hat{\Lambda}_0$ steps only at observed event times, so it is zero before the first event rather than carried backwards from it. The entry interval ends at the first event time, so carrying it backwards would give $\hat{\Lambda}_0(t_1) = \hat{\Lambda}_0(t_2)$ there, make $\Delta\hat{\Lambda}_0 = 0$ and set every prediction on that interval to zero, disregarding whatever the covariates said.

B.9 Partial likelihood on a held-out fold

The factor of (A.14) for customer i failing at t is

$$\frac{\exp(x_i'\hat{\beta})}{\sum_{l \in R(t)} \exp(x_l'\hat{\beta})},$$

with $R(t)$ the risk set. A held-out fold holds about a fifth of the customers, so its risk set is about a fifth the size, the denominator shrinks by roughly that factor, and each factor grows about fivefold. The fold's log partial likelihood rises by about $\log 5$ per event whether or not $\hat{\beta}$ is any better. A predicted hazard and its outcome score a row on their own, so the out-of-fold metrics carry no such dependence.

B.10 Likelihood ratio test for linearity

The statistic is twice the difference in maximised log-likelihood,

$$\chi^2_{\mathrm{LR}} = 2\big(\ell_{\text{spline}} - \ell_{\text{linear}}\big) \sim \chi^2_q, \tag{B.11}$$

where $q = 36$ is the number of extra parameters in the spline model [47]. The test is valid only if the linear model is a special case of the spline model. It is, since every straight line lies in the span of a natural spline basis.

Table B.4 repeats the test one covariate at a time. M1 is refitted with that covariate entered linearly and all others kept as splines, so $q = 2$.

Covariate	χ^2_2	p
Mean mileage ordinal	22.00	<0.001
Vehicle age SD	11.31	0.003
Usage concentration	10.71	0.005
Days since last new vehicle	6.22	0.045
Sessions (28d)	5.62	0.060
Scan proportion	4.86	0.088
Audi proportion	3.91	0.142
Volkswagen proportion	3.79	0.151
Clear proportion	3.63	0.163
Main-app drift	2.40	0.301
Onboarding delay	1.98	0.372
Actions-per-session drift	1.88	0.391
Gap SD (56d)	1.60	0.450
Actions per session (28d)	1.57	0.457
Gap mean (56d)	1.52	0.467
Session drift	1.17	0.557
Mean vehicle age	0.88	0.643
Main-app proportion	0.86	0.652

Table B.4: Per-covariate likelihood ratio tests. Each compares M1 with M1 refitted with that covariate entered linearly. The Bonferroni threshold for eighteen tests is $0.05/18 \approx 0.0028$.

B.11 Rescaling the M3 hazard ratio

For a proportion, one unit of x is its whole range. A realistic comparison uses a smaller move Δx, with $\beta(t) = \beta + \beta^{\mathrm{tt}} g(t)$ the M3 coefficient at time t,

$$\mathrm{HR}(\Delta x) = e^{\beta(t)\,\Delta x}. \tag{B.12}$$

With $\Delta x = \mathrm{sd}_x = 0.18$, every log hazard ratio is multiplied by 0.18, so it is more than five times smaller but keeps its sign. The hazard ratio therefore still crosses 1 wherever $\beta(t)$ changes sign. For the one-click proportion under $g(t) = t$, $\beta(56) = -0.648$ and $\beta(798) = 0.556$, so the hazard ratio per SD runs from $e^{0.18\times(-0.648)} = 0.890$ to $e^{0.18\times 0.556} = 1.105$.

C

Covariate Specification

Mileage buckets are ordinal-encoded in low-to-high order of their lower edges, with unknowns coded as -1, and make is one-hot encoded per vehicle. Table C.1 lists every covariate slot.

Built (33)	Fitted (27)	Dropped
n_sessions_0_28	n_sessions_0_28	
sessions_drift_0_28_vs_28_56	sessions_drift_0_28_vs_28_56	
actions_per_session_0_28	actions_per_session_0_28	
actions_per_session_drift_0_28_vs_28_56	actions_per_session_drift_0_28_vs_28_56	
mean_gap_0_56	mean_gap_0_56	
sd_gap_0_56	sd_gap_0_56	
CV_gap_0_56		Identity
active_flag_0_56	active_flag_0_56	
recency		Collinear
account_tenure		Identity
onboarding_delay	onboarding_delay	
weekend_share_0_28	weekend_share_0_28	
usage_concentration	usage_concentration	
prop_scan_0_56	prop_scan_0_56	
prop_coding_0_56	prop_coding_0_56	
prop_oca_0_56	prop_oca_0_56	
prop_clear_0_56	prop_clear_0_56	
prop_history_screen_0_56	prop_history_screen_0_56	
prop_live_data_0_56	prop_live_data_0_56	
prop_other_0_56		Reference
prop_main_0_56	prop_main_0_56	
prop_main_drift_0_28_vs_28_56	prop_main_drift_0_28_vs_28_56	
prop_vag_0_56		Reference
prop_make_volkswagen	prop_make_volkswagen	
prop_make_audi	prop_make_audi	
prop_make_skoda	prop_make_skoda	
prop_make_other_rare		Reference
vehicle_mean_age	vehicle_mean_age	
vehicle_age_sd	vehicle_age_sd	
vehicle_mileage_ordinal	vehicle_mileage_ordinal	
prop_in_prod	prop_in_prod	
n_new_vehicles_0_28	n_new_vehicles_0_28	
days_since_last_new_vehicle	days_since_last_new_vehicle	

Table C.1: Every covariate slot, grouped as in Figure 3.5. The first column is what feature engineering produces and the second what enters the model, so a gap marks a covariate that does not. Names drop the trailing `_days` and the `intensity_` and `overall_prop_vehicle_` prefixes, which are constant across the set. *Reference* is an omitted category of a block whose shares sum to one. *Identity* is an exact linear function of covariates already present, and *Collinear* is redundancy without an exact identity.